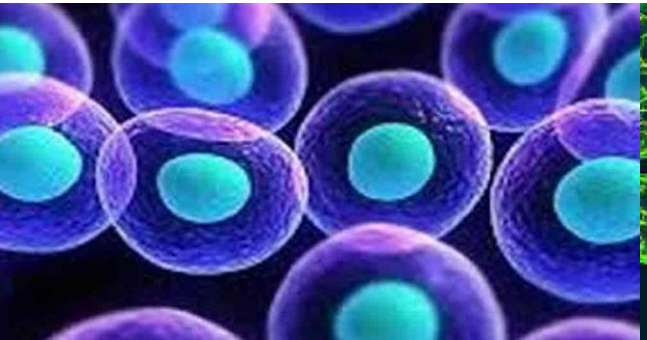
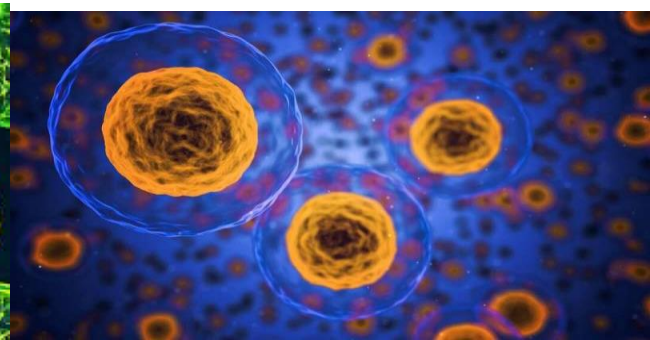
Part 2

The chemicals components of the cells





Class	Contents	Hours		
		Lect.	Exr.	Prc.
1	[1] Topic 1: The fundamental unit of life	03		
2	[2] Topic 2: The chemical components of the cells	03		
3	[3] Topic 3: Eukaryotic cell structure	12		
4	[4] Topic 6: Cell cycle and cell communication	03		
5	[5] Topic 8: Apoptosis, autophagy, cancer, and stem cell	03		
6	[6] Topic 7: Cell metabolism	02		
7	[7] Topic 9: Prokaryotic cell and virus	02		
8	[8] Topic 10: Cell techniques and their applications	02		
9	[9] Topic 11: Lab work			06
10	Tutor			
	Total:	30	05	06

Essential concepts

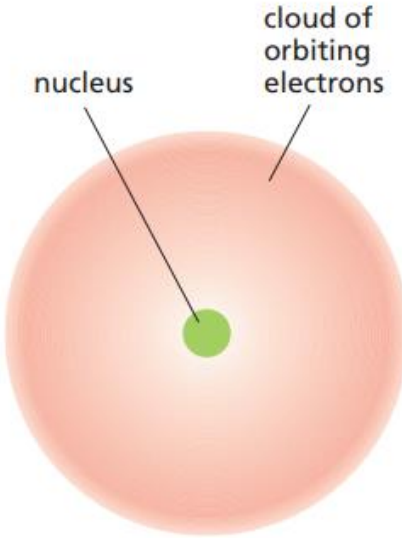
- Chemical bonds
- Small molecules in cells
- Macromolecules in cells

Chemical bonds

What makes up matter in the world?

Have you've ever seen a Periodic Table of Elements?

What is the smallest particle of an element that still retains its distinctive chemical properties



Matter is made of combinations of *elements*

The smallest particle of an element that still retains its distinctive chemical properties is an *atom*

The characteristics of substances other than pure elements—including the materials from which living cells are made—depend on which atoms they contain and the way these atoms are linked together in groups to form molecules

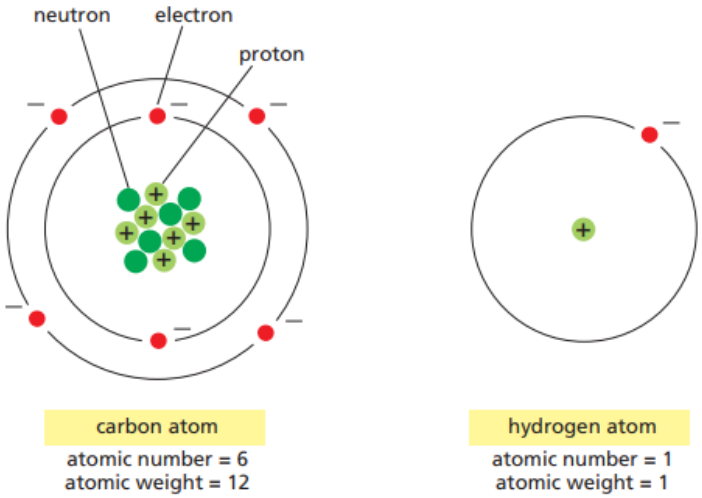
-> To understand living organisms: Crucial to know the chemical bonds that hold atoms together in molecules are formed

An atom consists of a nucleus surrounded by an electron cloud.
The dense, positively charged **nucleus** contains most of the atom's mass. Electrons occupy space around: much lighter and negative charge, governed by the laws of quantum mechanics.

- Electron: negatively charged
- Protons: positively charged
- Neutron: electrically neutral

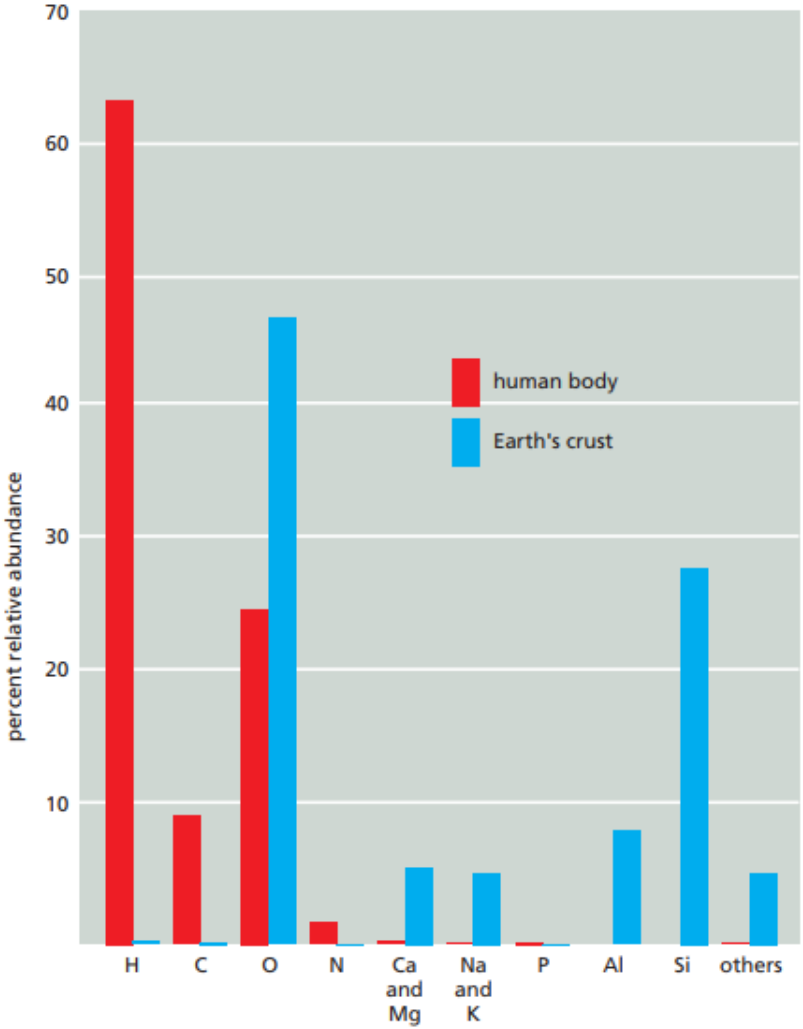
Neutrons have essentially the same mass as protons. They contribute to the structural stability of the nucleus - if there are too many or too few, the nucleus may disintegrate by radioactive decay—but they do not alter the chemical properties of the atom

≠ Neutron: Isotopes

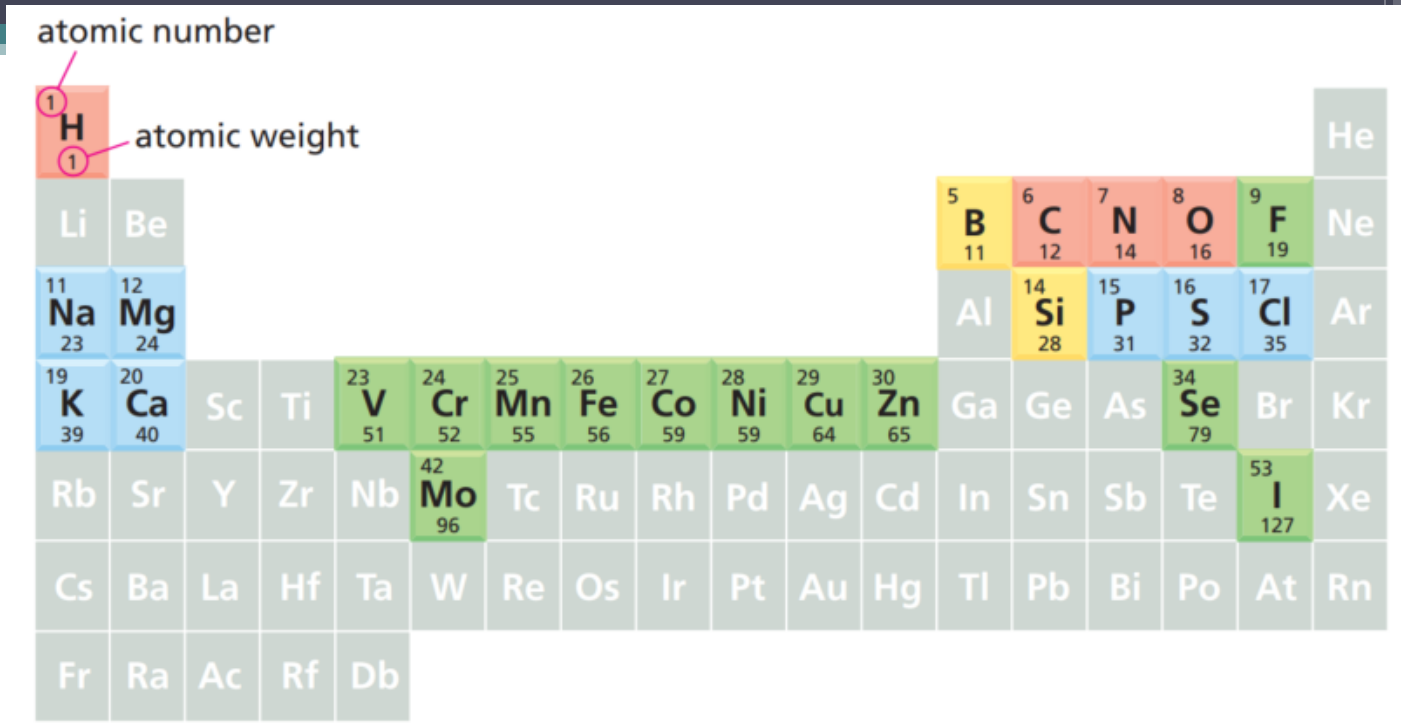


Schematic representations of an atom of carbon and an atom hydrogen

Cells Are Made of Relatively Few Types of Atoms



The distribution of elements in the Earth's crust differs radically from that in a living organism



The chemistry of life is predominantly the chemistry of lighter elements.

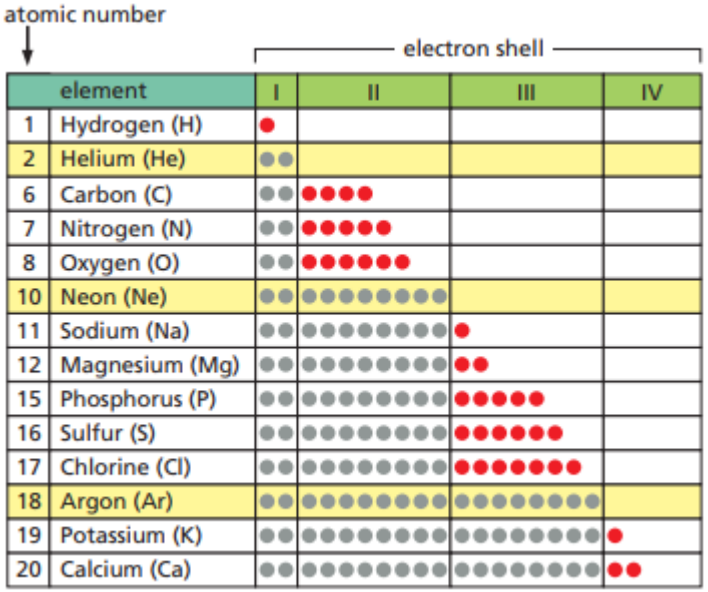
Red: constitute 99% of the total number of atoms present in the human body and about 96% of our total weight.
Blue: 0.9% of the total number of atoms.
Green: are required in trace amounts by humans.
Yellow: unclear whether those elements are essential in humans or not.

The Outermost Electrons Determine How Atoms Interact

Incompletely filled electron shell → less stable → strong tendency to interact with other atoms so as to either gain or lose enough electrons to achieve a completed outermost shell.

-> This electron exchange can be achieved either by transferring electrons from one atom to another or by sharing electrons between two atoms

→ 2 types of chemical bonds



All of the elements commonly found in living organisms have outermost shells that are not completely filled with electrons (red) and can thus participate in chemical reactions with other atoms. Inert gases (yellow), in contrast, have completely filled outermost shells and are thus chemically unreactive

Covalent Bonds

❖ Strong forces form a **covalent bond** when two atoms share one pair of electrons ("single" bond) or multiple pairs of electrons ("double" bond, "triple" bond, etc.)

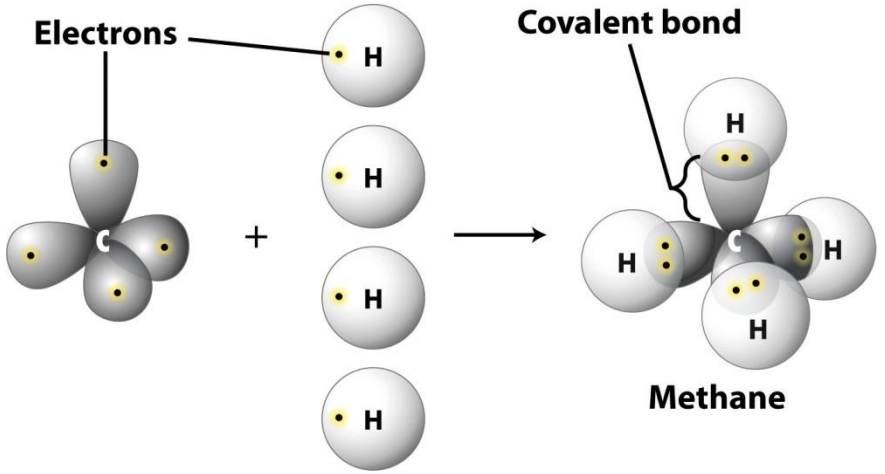


Figure 2-2
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Covalent bonds form by the sharing of electron

Covalent bonds, the strong forces that hold atoms together into molecules, form when atoms share electrons from their outer most electron orbitals. Each atom forms a defined number and geometry of covalent bonds

Covalent Bonds

1. The Electronic Structure of an Atom Determines the Number and Geometry of Covalent Bonds lt Can Make

❖ All the biological building blocks are organized around the carbon atom

TABLE 2-1 Bonding Properties of Atoms Most Abundant in Biomolecules

ATOM AND OUTER ELECTRONS	USUAL NUMBER OF COVALENT BONDS	TYPICAL BOND GEOMETRY
H	1	H
O	2	O
S	2, 4, or 6	S
N	3 or 4	N
P	5	P
C	4	C

Table 2-1
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Covalent Bonds

Electrons May Be Shared Equally or Unequally in Covalent Bond

❖The extent of an atom's ability to attract an electronic called its ***electronegativity*** (affinities for electrons)

❖**Non-polar**: the bonding electrons are shared equally between the two atoms

Ex: in the case for most C-C; C-H bonds

❖**Polar**: bonded atoms have different electronegativities → unequal sharing of the electrons

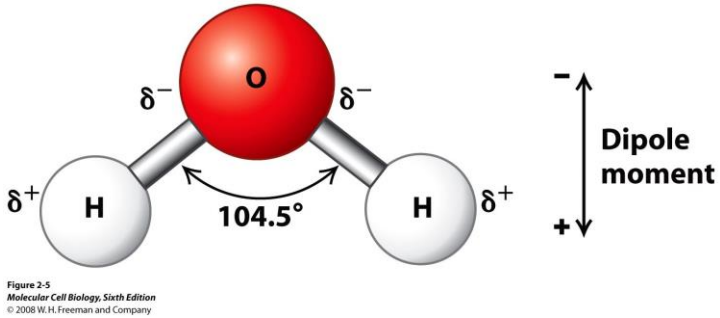
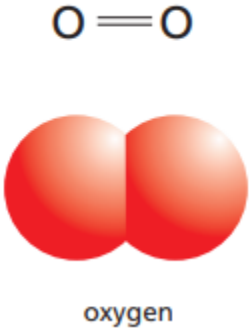


Figure 2-5
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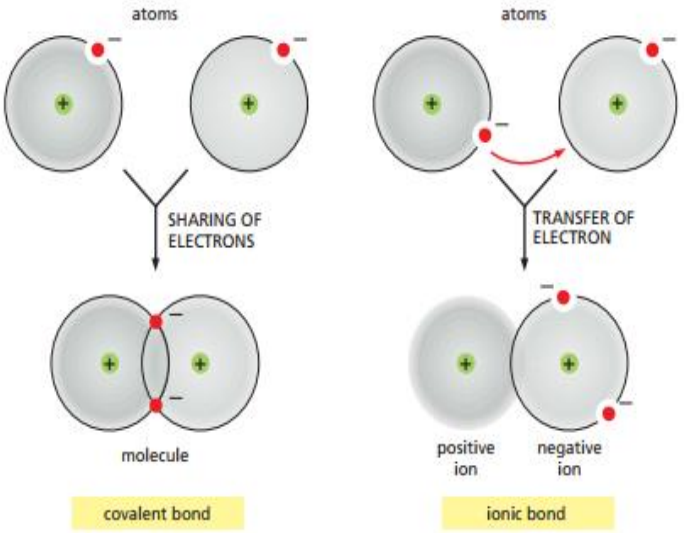
Because of the difference in the electronegativities of H and O, each of the polar H-O bonds in water is a dipole. electrons are more strongly attracted to the oxygen nucleus than to the H nucleus, as indicated by the distributions of the partial negative (δ–) and partial positive (δ+) charges

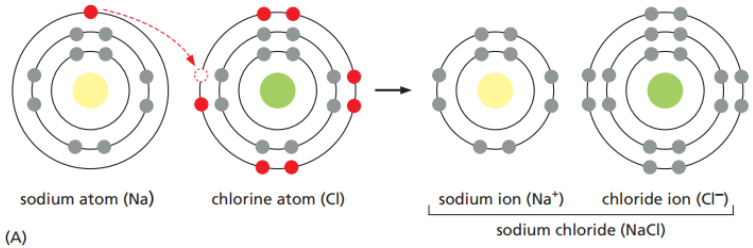
Covalent bonds

Noncovalent Bonds – Ionic interactions

Ionic Bonds Form by the Gain and Loss of Electrons

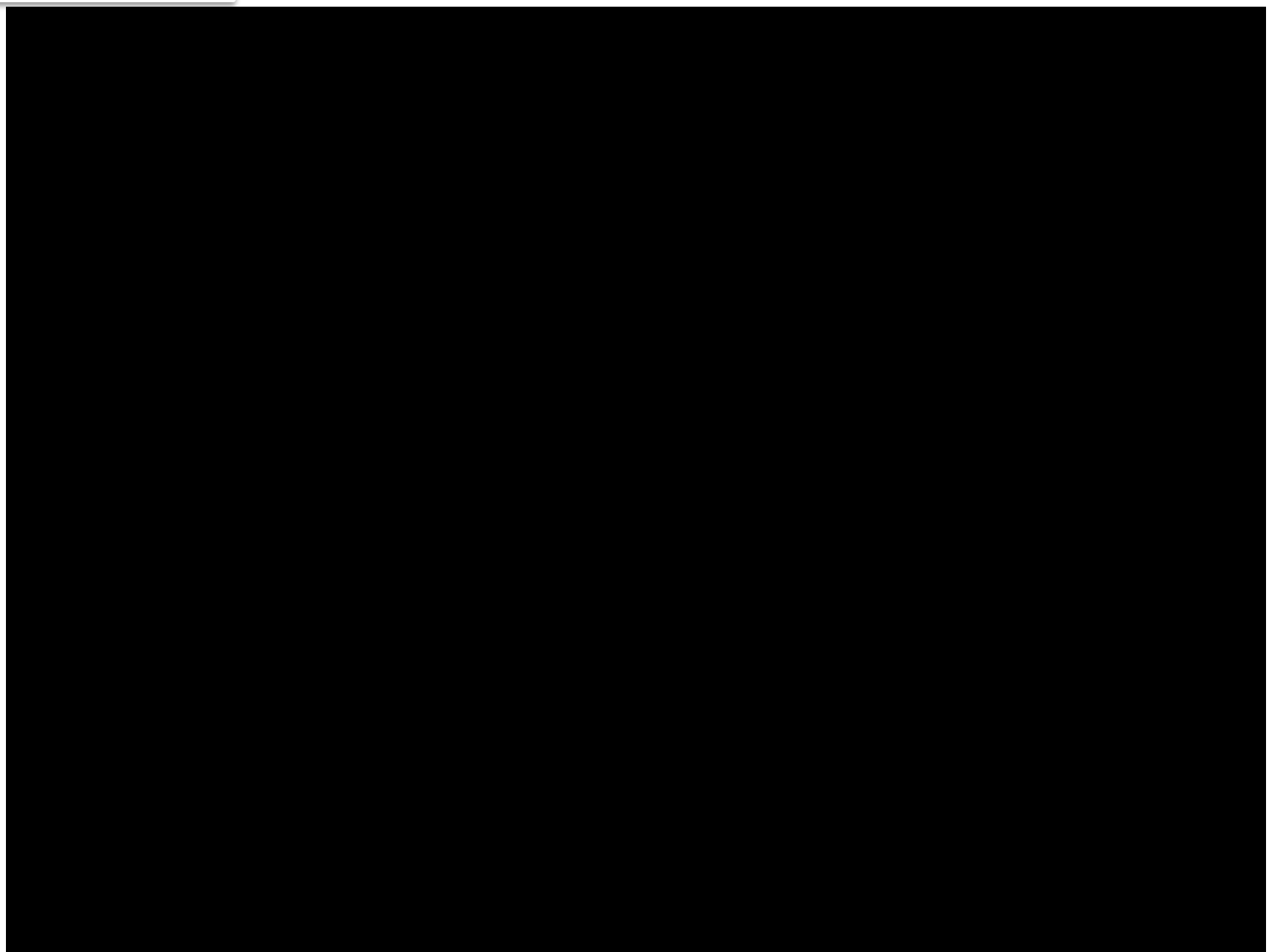
Result from the attraction of a positively charged ion-a cation-for a negatively charged ion-an anion





In sodium chloride (NaCl), for example, the bonding electron contributed by the sodium atom is completely transferred to the chlorine atom

Ionic Bonds



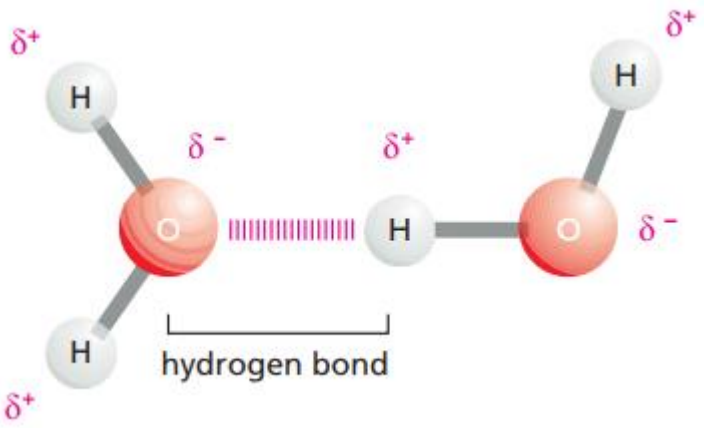
Noncovalent Bonds – Hydrogen bonds

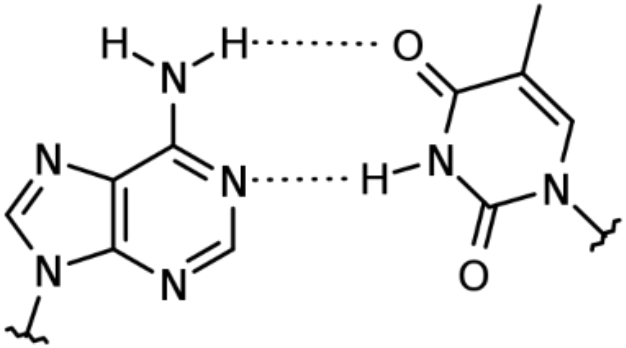
The interaction of a partially positively charged hydrogen atom in a molecular dipole (e.9.,water) with unpaired electrons from another atom, either in the same (intramolecular) or different (intermolecular) molecule

$$D^{\delta-}—H^{\delta+} + :A^{\delta-} \rightleftharpoons D^{\delta-}—H^{\delta+}\cdots\cdots:A^{\delta-}$$

Hydrogen bond

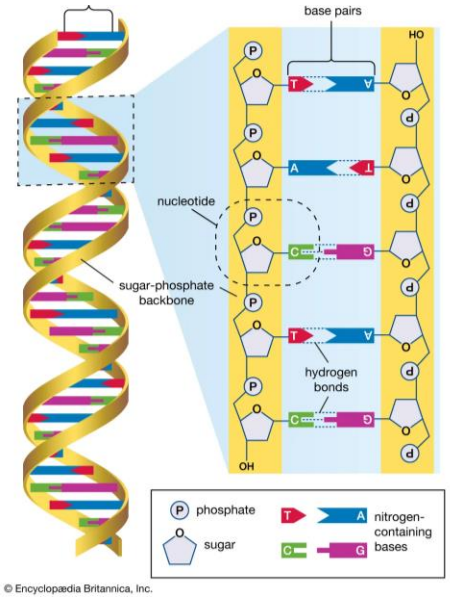
A hydrogen bond can form between two water molecules. These bonds are largely responsible for water's life-sustaining properties—including its ability to exist as a liquid at the temperatures inside the typical mammalian body.





Adenine Thymine

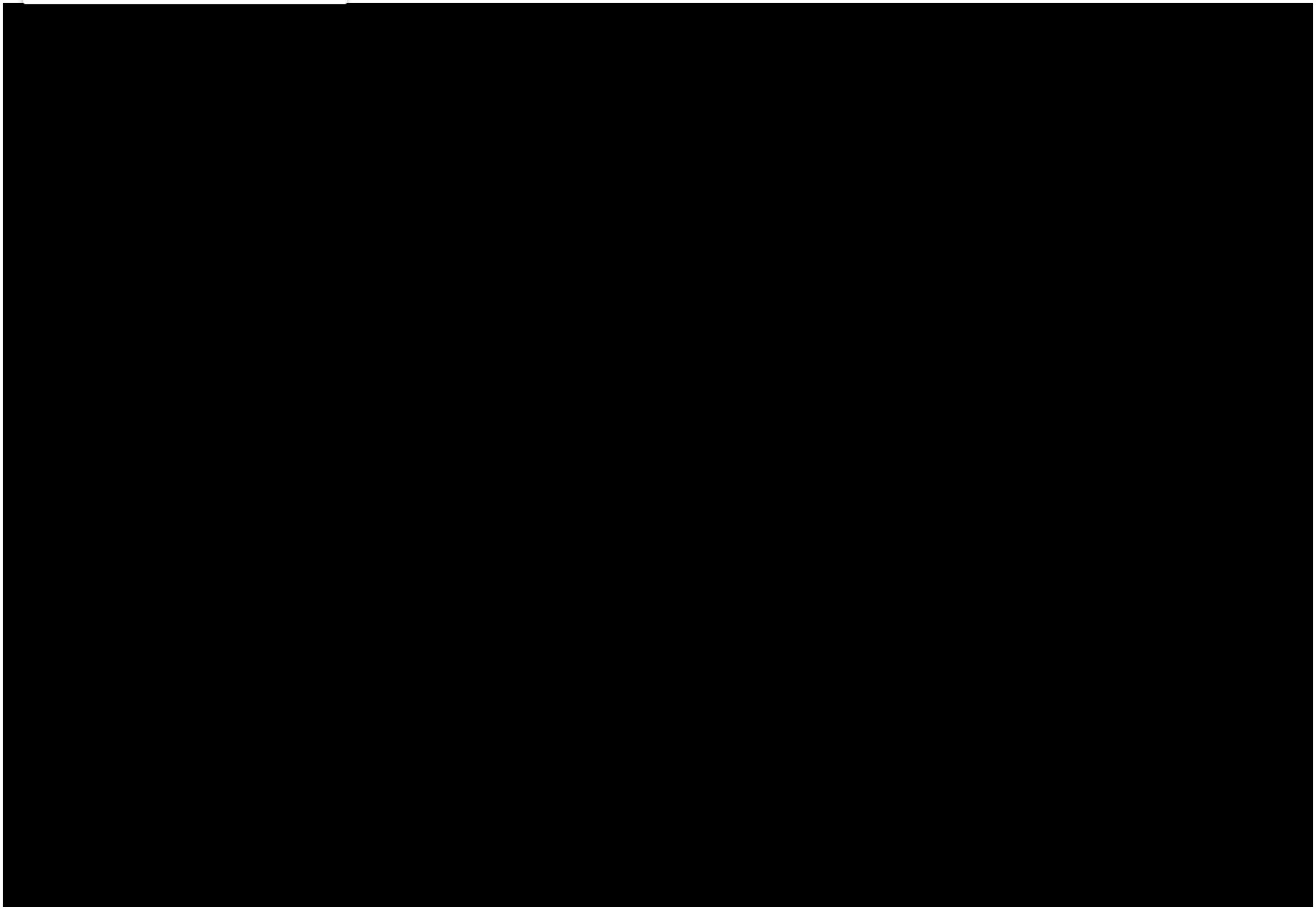
Hydrogen bonds can form between molecules besides water
+ Oxygen to hydrogen
+ Nitrogen to hydrogen



Hydrogen bonds are essential

+ Nucleic acids
+ Proteins

Hydrogen bonds



Noncovalent Bonds – Van der Waals interactions

When any two atoms approach each other closely they create a weak, non specific attractive force called a Van der Waals interaction

Noncovalent Bonds – Hydrophobic interactions

In an aqueous environment, **nonpolar molecules** or **nonpolar portions** of larger molecules are driven together by the **hydrophobic effect**, thereby reducing the extent of their direct contact with water molecule

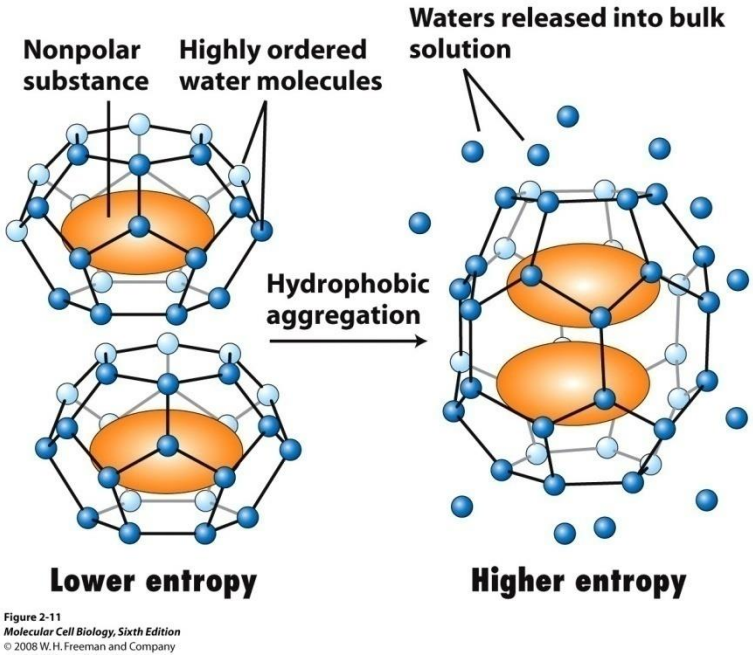


Figure 2-11
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Schematic depiction of the hydrophobic effect

Bonds strength

Covalent Bonds Are Much Stronger and More Stable Than Non covalent Interaction

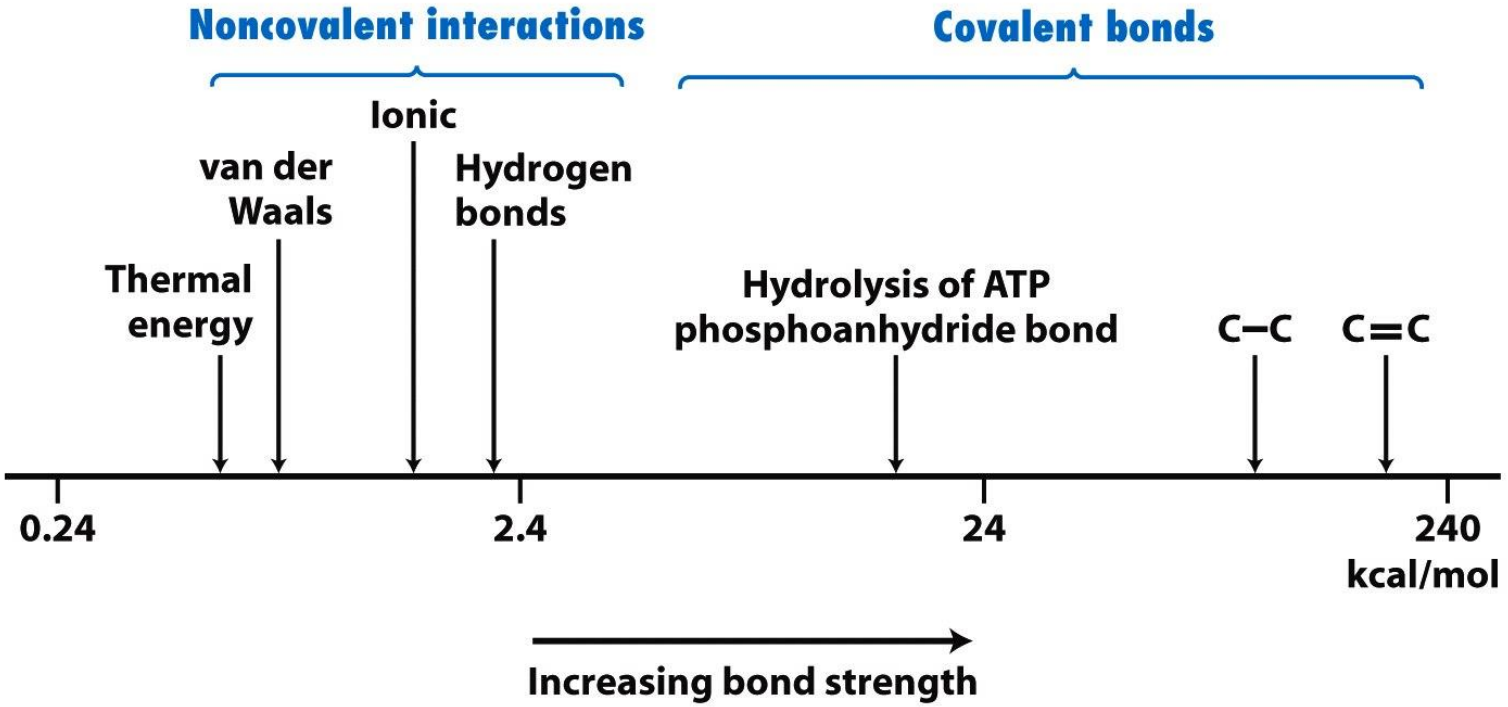


Figure 2-6
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Relative energies of covalent bonds and non-covalent interactions

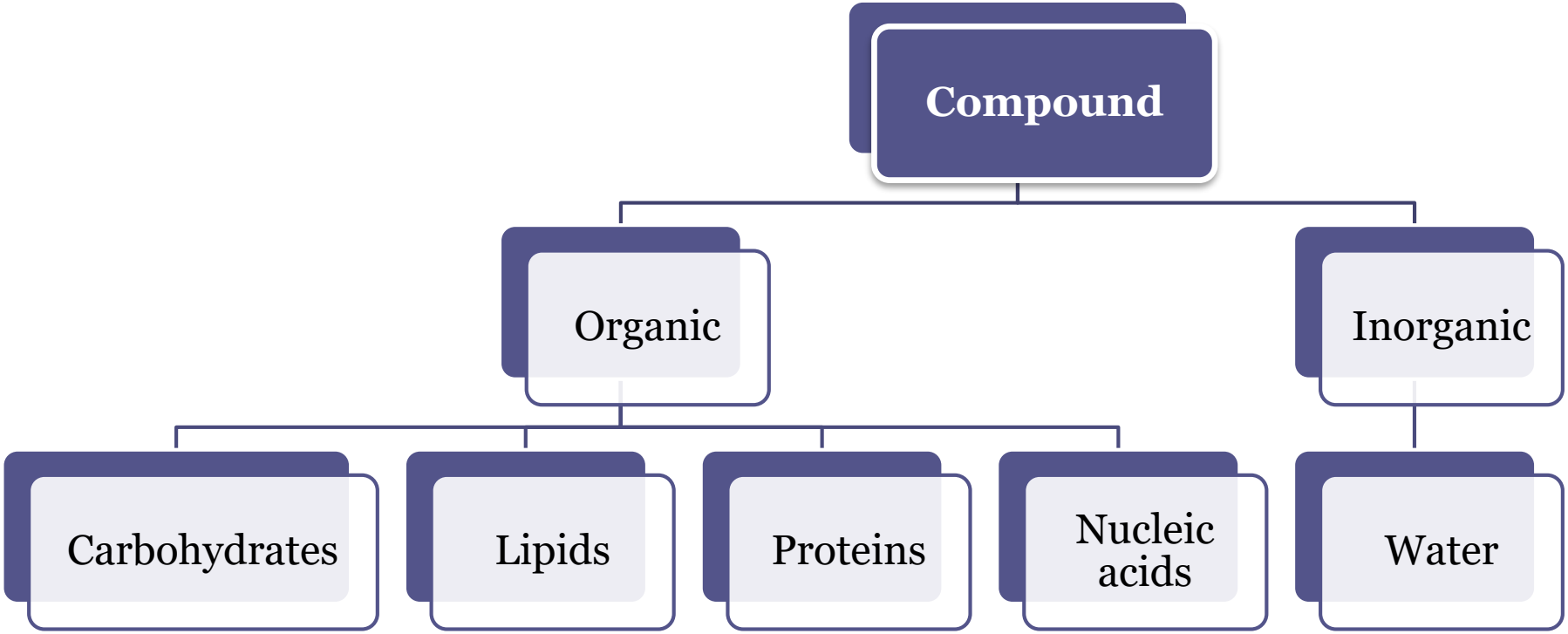
Small molecules in cells

92 element occurring naturally in nature

25 element are needed to build living organisms

Main element (CHON) are the most frequently found elements in cells, forming about 96% of the human body mass

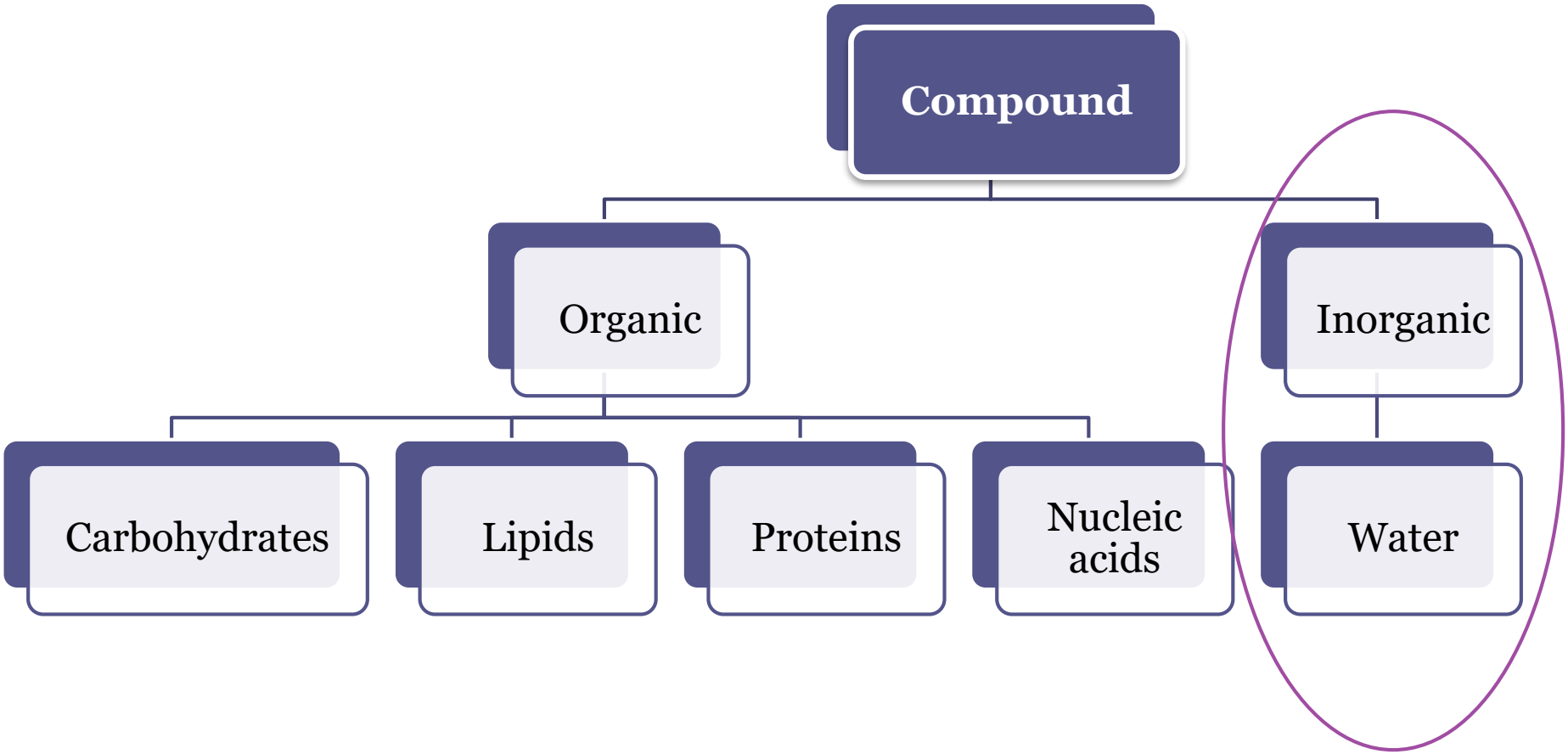
Trace-elements are the elements are found in small quantity in cells, but are important in biological processes


Compound
Organic
Inorganic
Carbohydrates
Lipids
Proteins
Nucleic acids
Water

Approximate Chemical Composition of a Rapidly Dividing Cell (*E. coli*)

Material	% Total Wet Wt.	Different Kinds of Molecules/Cell
Water	70	1
Nucleic acids		
DNA	1	1
RNA	6	
Ribosomal		3
Transfer		40
Messenger		1000
Nucleotides and metabolites	0.8	200
Proteins	15	2000-3000
Amino acids and metabolites	0.8	100
Polysaccharides	3	200
(Carbohydrates and metabolites)		
Lipids and metabolites	2	50
Inorganic ions	1	20
(Major minerals and trace elements)		
Others	0.4	200
	100	

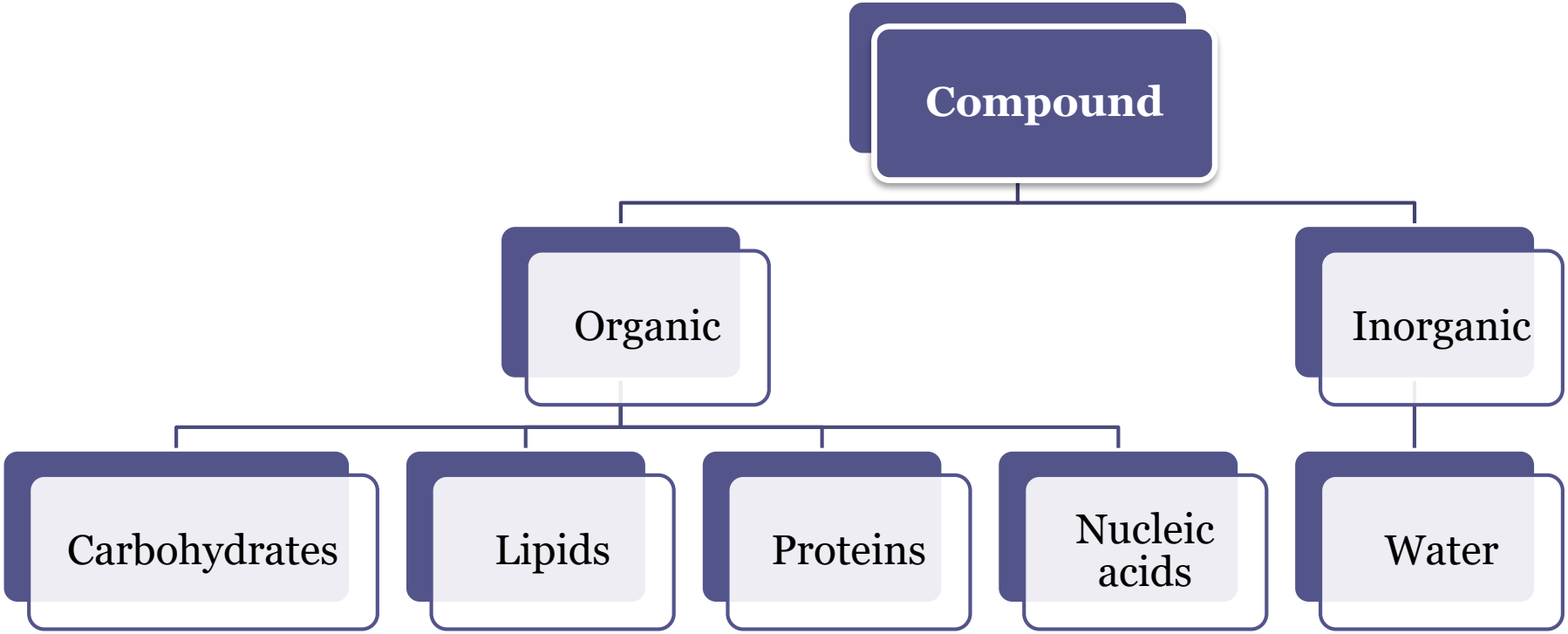
Data from Watson JD: Molecular Biology of the Gene, 2nd ed., Philadelphia, PA: Saunders, 1972.


Compound
Organic
Inorganic
Carbohydrates
Lipids
Proteins
Nucleic acids
Water

Inorganic compound - Water

- The most abundant substance in the cell accounts for about 70% of a cell's weight
- Most intracellular reactions occur in an aqueous environment
- Where did it come from?

  - condensation from the primary atmosphere
  - release of gases from the Earth interior
  - extraterrestrial origin


Compound
Organic
Inorganic
Carbohydrates
Lipids
Proteins
Nucleic acids
Water

Organic compound

Four main types of small organic molecules:

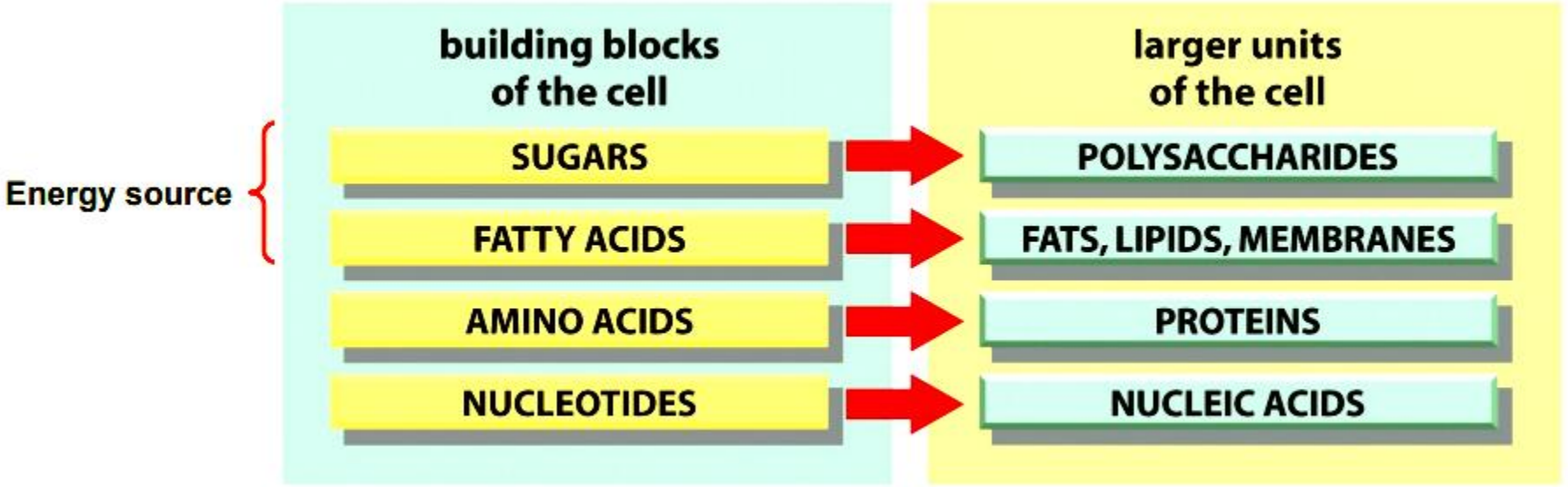
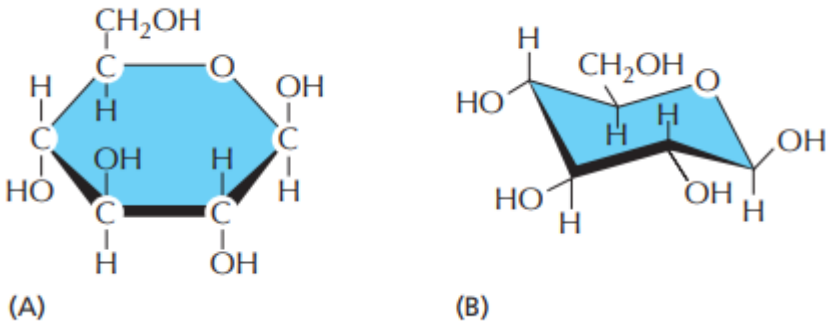


Figure 2-17 Molecular Biology of the Cell (© Garland Science 2008)

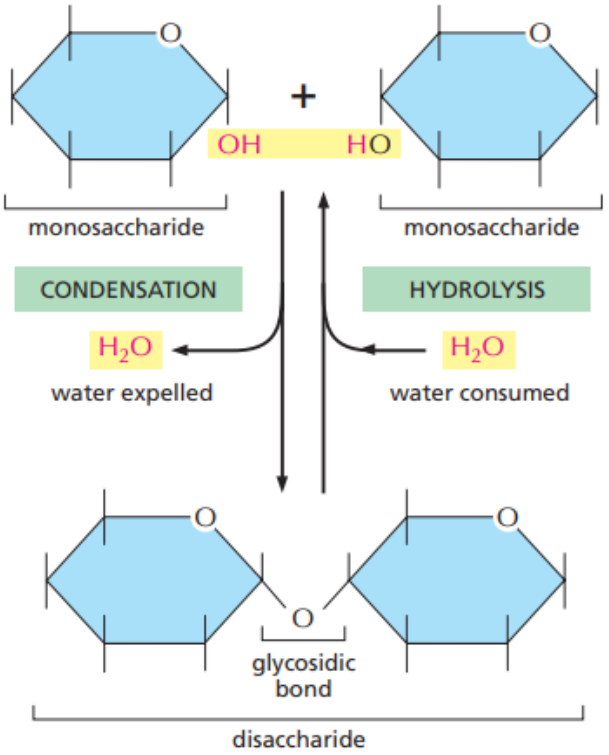
Carbohydrates

The simplest sugars—the monosaccharides—are compounds with the general formula $(CH_2O)_n$, where n is usually 3, 4, 5, or 6

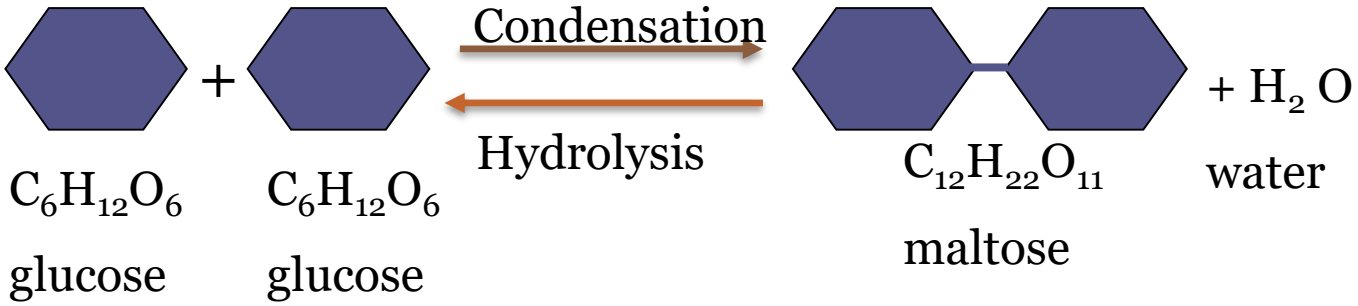
Sugars, and the larger molecules made from them, are also called **carbohydrates**

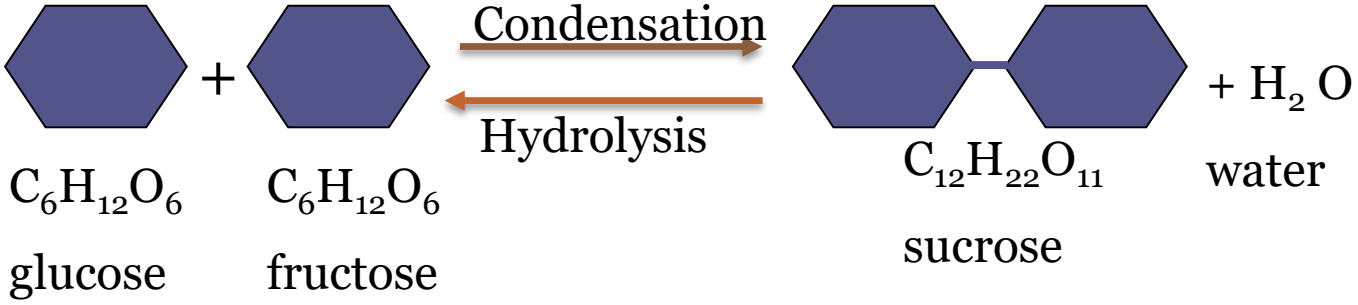


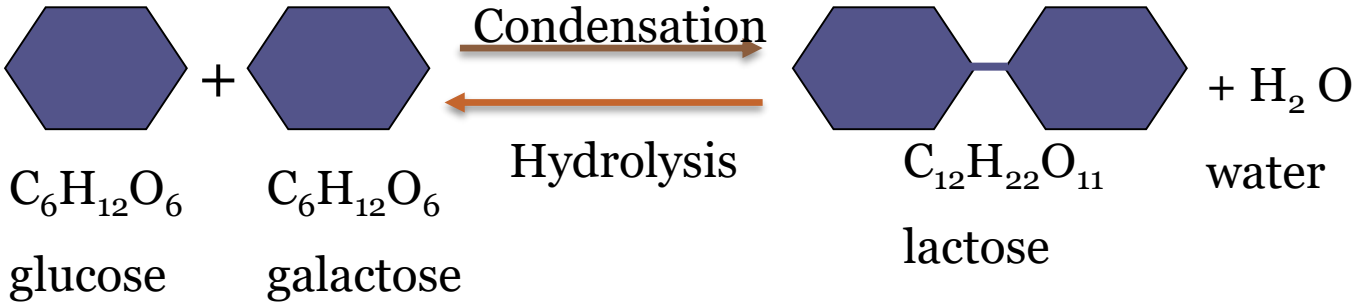
The structure of glucose, a monosaccharide, can be represented in several ways



Disaccharides







Polysaccharides

❖ Polymers of simple sugars (monosaccharides)

- **Homogeneous polymers**: contain only one kind of sugar/ considered to be ‘noninformational”

- **Heterogeneous polymers**: contain 8-10 types of sugars

Glycogen – in humans and animals
Starch and cellulose – in plants

Functions

❖**Energy sources**

-**glucose** – key energy source for cells → in a series of reactions is broken down to smaller molecules to release the energy - cells use simple polysaccharides

-composed only of glucose units → **glycogen** (animal cells) and **starch** (plant cells)

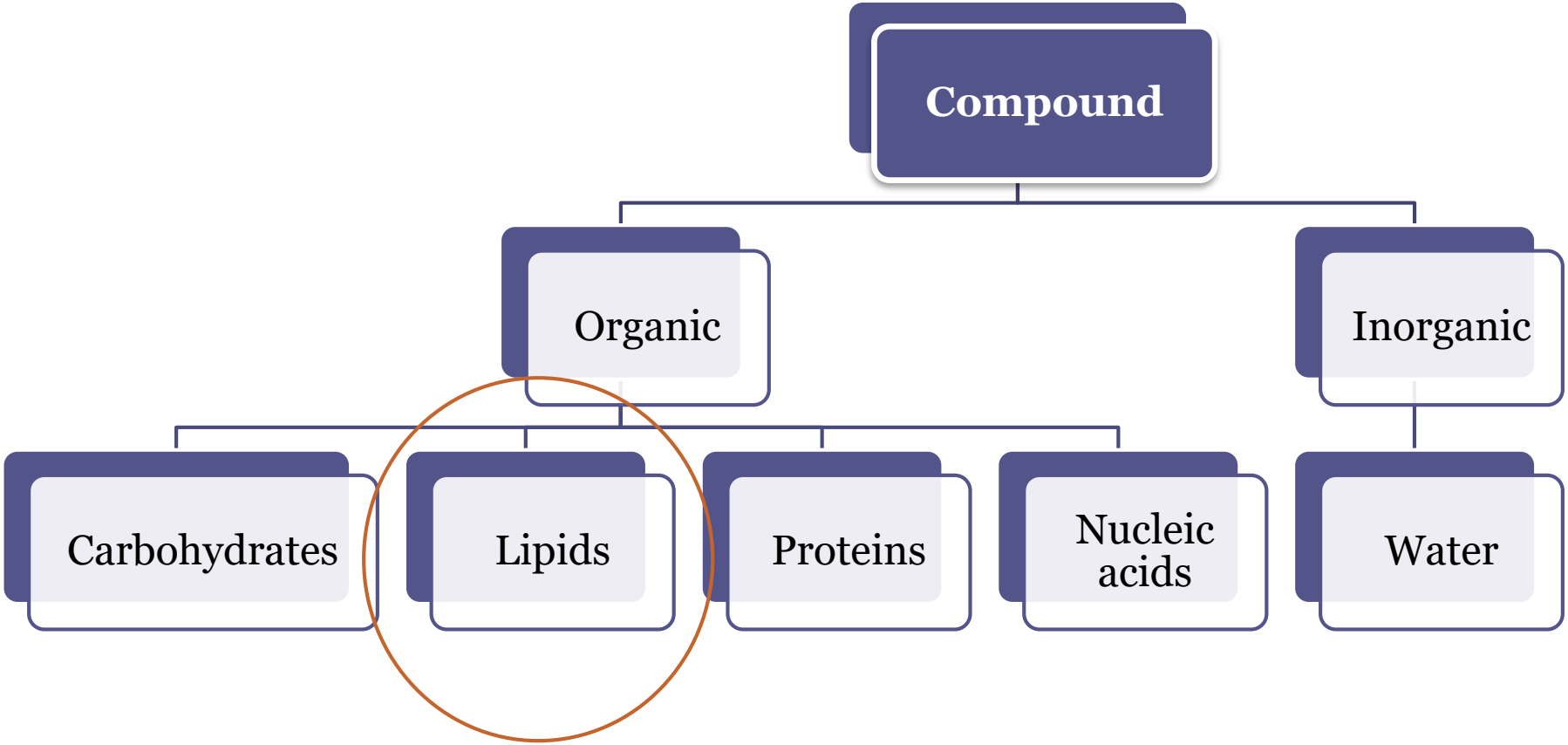
❖ **Structure support**

- **cellulose** – glucose polysaccharide (the most abundant organic chemical on Earth!)

- **chitin** – linear polymer of N-acetylglucosamine insect exoskeletons and fungal cell walls

❖ **Molecular markers**

- **glycoproteins and glycolipids** of the cell membrane → selective recognition by other cells, help cells adhere to one another, Blood type…


Compound
Organic
Inorganic
Carbohydrates
Lipids
Proteins
Nucleic acids
Water

Lipids

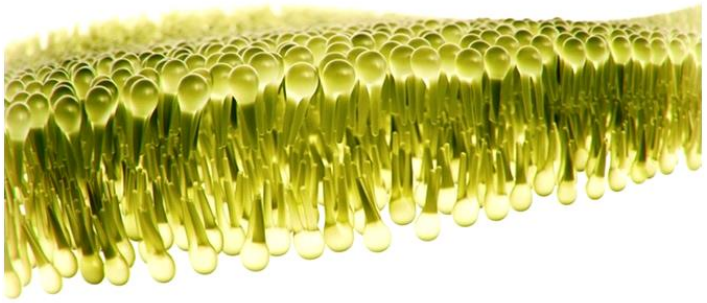
❖ From the Greek word lipos, meaning "fat"

❖ They are non-polar; not soluble in water; soluble in non-polar solvents such as chloroform

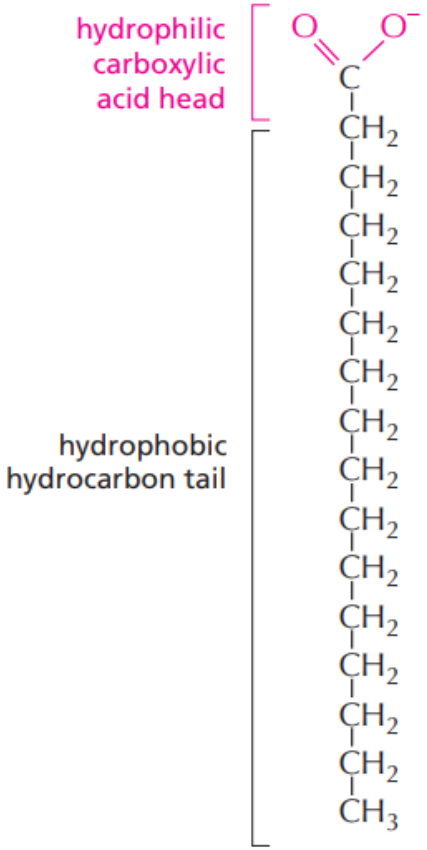
❖ Two different structural parts:

- Hydrophilic head: chemically active

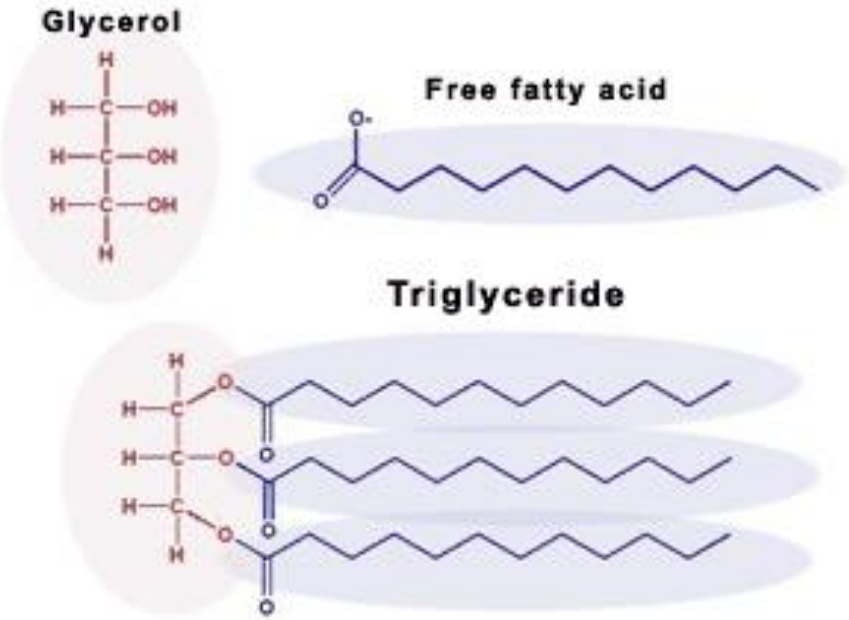
- Hydrophobic tail: differences between hydrocarbon chains- not very active chemical



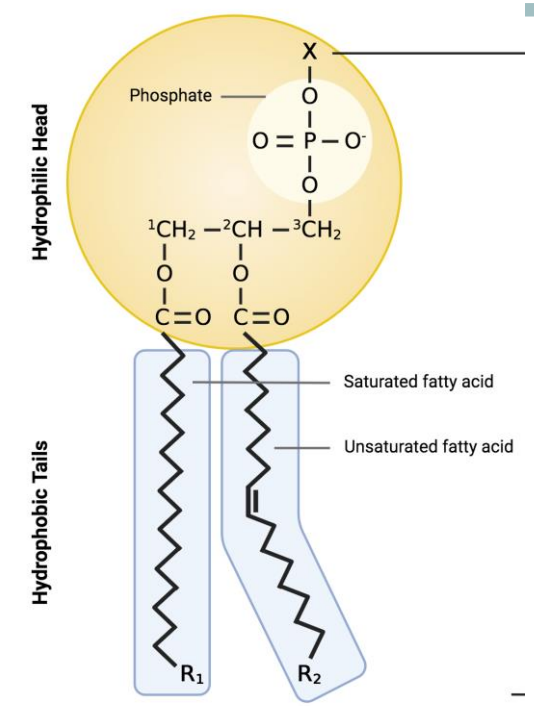
Human lipid bilayer - 3D Rendering. Image Credit: Crevis / Shutterstock



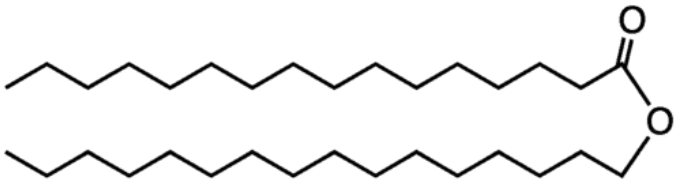
Fatty acids have both hydrophobic (kỵ nước) and hydrophilic (ưa nước) components.



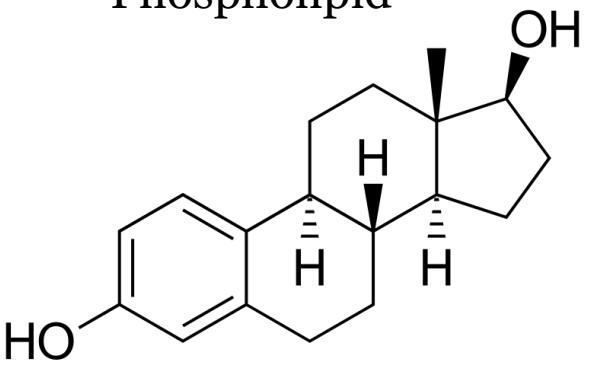
Fat (triglyceride)



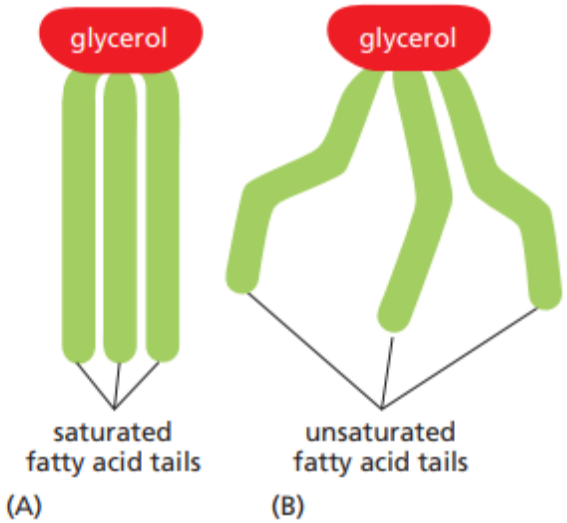
Phospholipid



Wax



Steroids



The properties of fats depend on the length and saturation of the fatty acid chains they carry

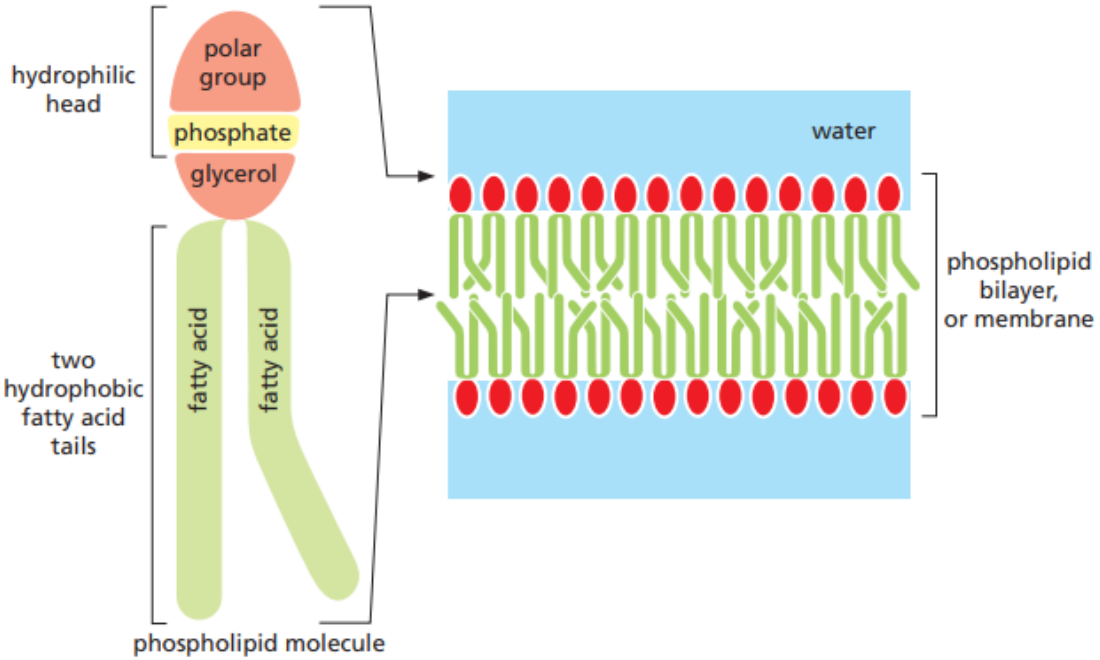
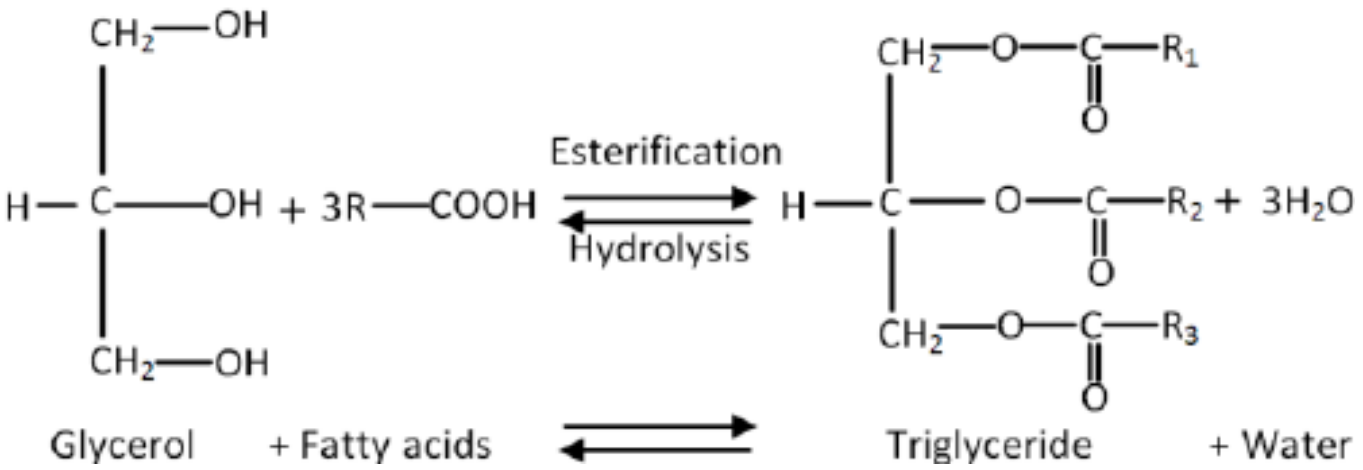


Figure 2–21 Phospholipids can aggregate to form cell membranes. Phospholipids are composed of two hydrophobic fatty acid tails joined to a hydrophilic head. In an aqueous environment, the hydrophobic tails pack together to exclude water, forming a lipid bilayer, with the hydrophilic heads of the phospholipid molecules on the outside, facing the aqueous environment, and the hydrophobic tails on the inside.


CH2—OH
H—C—OH + 3R—COOH
CH2—OH
Esterification
Hydrolysis
CH2—O—C—R1
H—C—O—C—R2 + 3H2O
CH2—O—C—R3
O
Glycerol + Fatty acids
Triglyceride + Water

Lipids

❖ Energy source

Stored in the cytoplasm of many cells in the form of droplets of triacylglycerol molecules (produce about **six times** as much usable energy, weight for weight, as glucose)

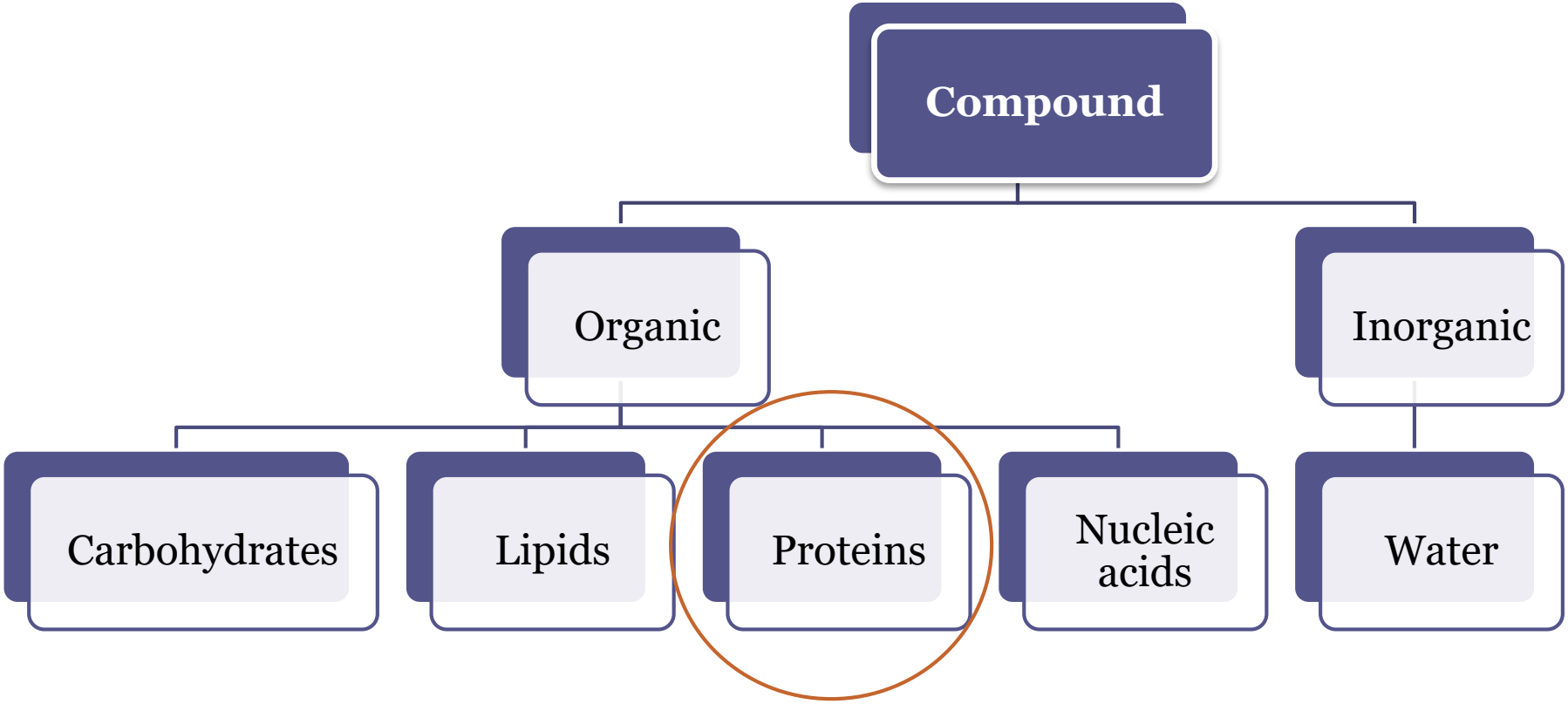
- animal fats (meat, butter and cream)
- plant oils (corn and olive oil)

❖ Waterproofing (waxes)

❖ Construction of biological membranes

- Phospholipids
- Glycolipids
- Cholesterol

❖ **Signaling**: Steroid hormones like testosterone, and estrogens


Compound
Organic
Inorganic
Carbohydrates
Lipids
Proteins
Nucleic acids
Water

Proteins - Amino acids

❖ All have carboxylic acid group (COOH) and amino group (NH2) both linked to a single C-atom (α-carbon)

❖ Differing Only in Their Side Chains Compose Proteins

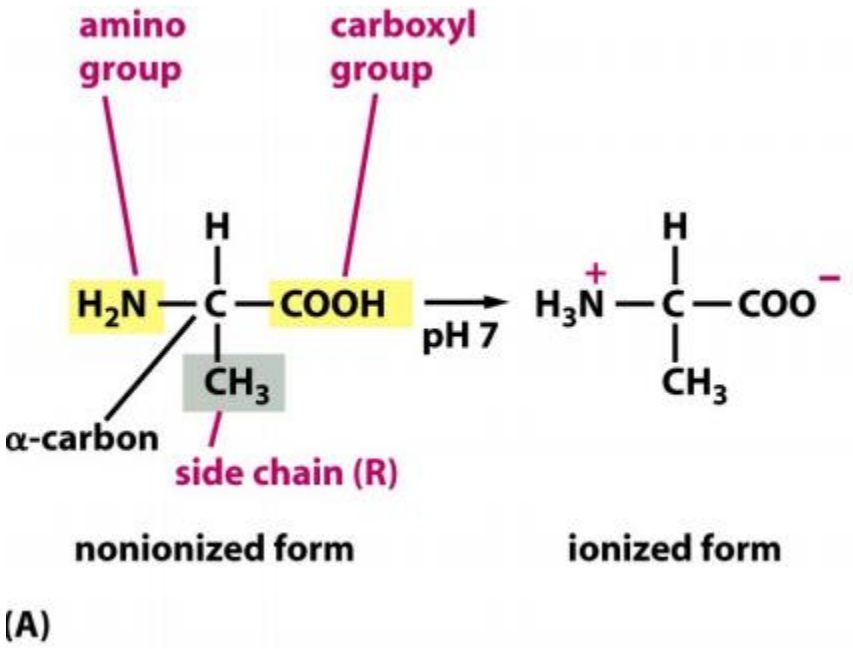
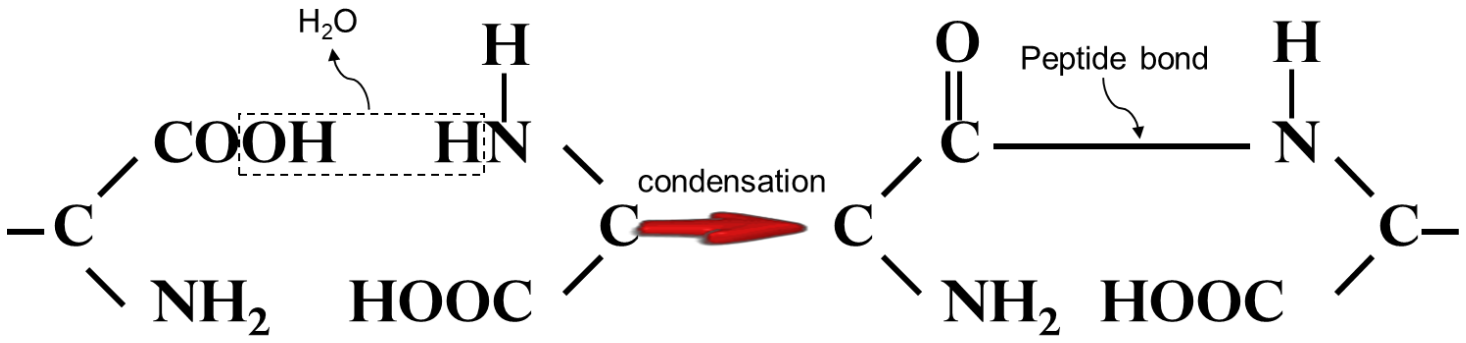


Figure 2-23 Molecular Biology of the Cell (© Garland Science 2008)

Two amino acids can combine together to form a **dipeptide** by a **condensation reaction** between the carboxyl group of one and the amino group of the other. The resulting a bond liking the two amino acids that is called a **peptide bond.**

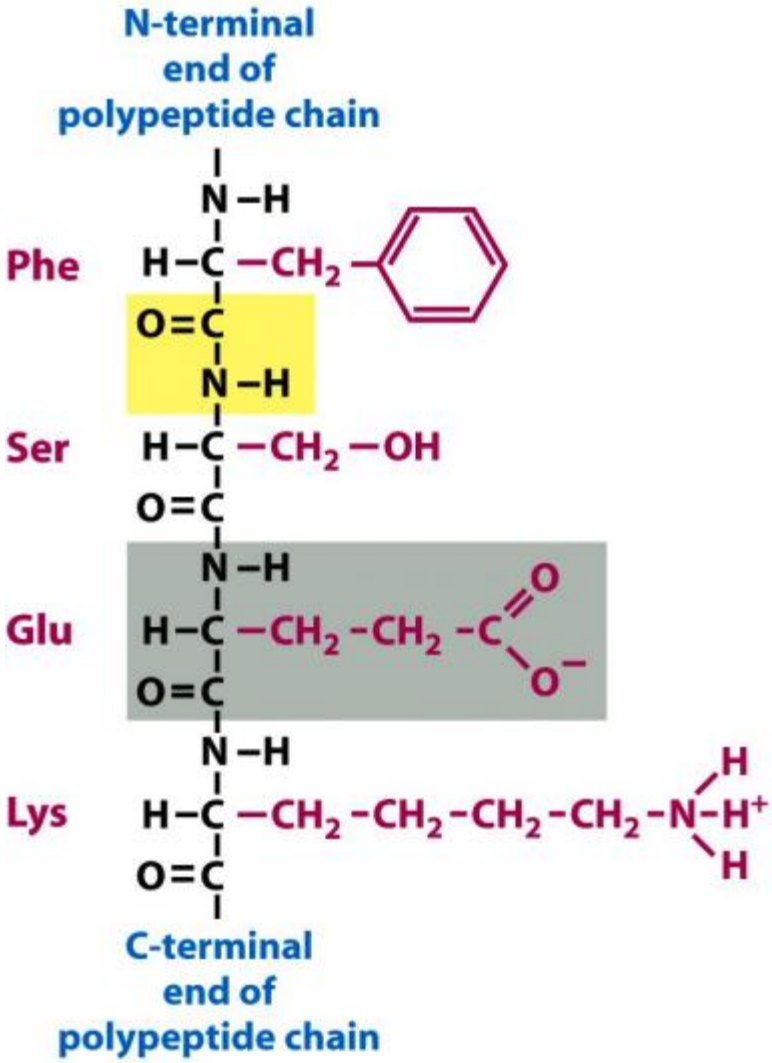


Proteins - Amino acids

❖ Amino acids linked together → peptide bond

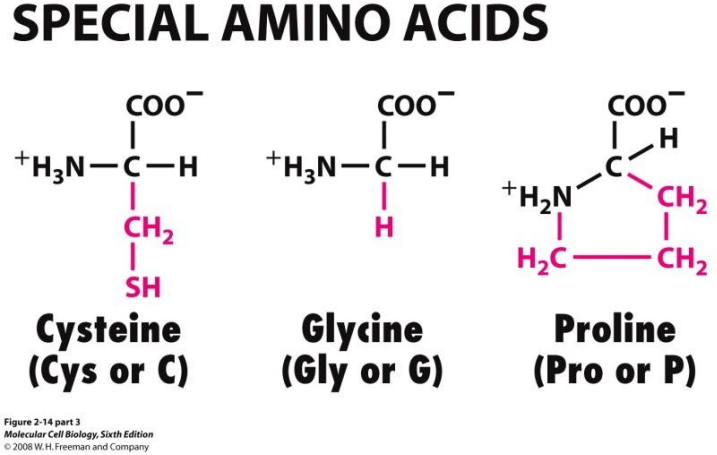
❖ Polypeptide or protein –> two chemically distinct ends:

-NH2 (N-terminus)
-COOH (C-terminus)



Special amino acids

Cysteine contains a thiol (-SH) group and can form covalent bonds with other cysteines.



Proline has an R group that's linked back to its own amino group, forming a ring structure. This makes it an exception to the typical structure of an amino acid, since it no longer has the standard $NH3^{=}$ amino group→ very rigid

The smallest amino acid**, glycine**, has a single hydrogen atom as its R group. Its small size allows it to fit into tight space

Hierarchical Structure of Protein

1. **Primary structure**: the linear sequence of amino acids linked together by peptide bonds

2. **Secondary structure**: folding of the polypeptide chain into local α helices or β sheets

3. **Tertiary structure:** 2nd structure with various loops and turns in a single polypeptide chain pack into a larger independently stable structure

4. **Quaternary structure**: some individual polypeptide with their own tertiary structures can associate into a quaternary structure – multichain complex

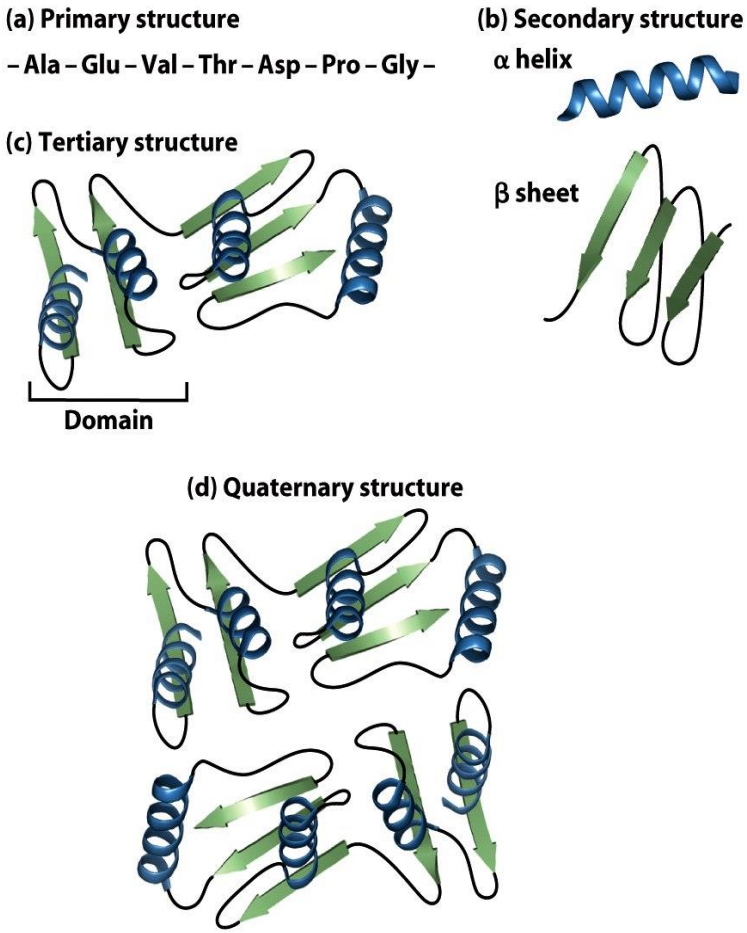


Figure 3-2
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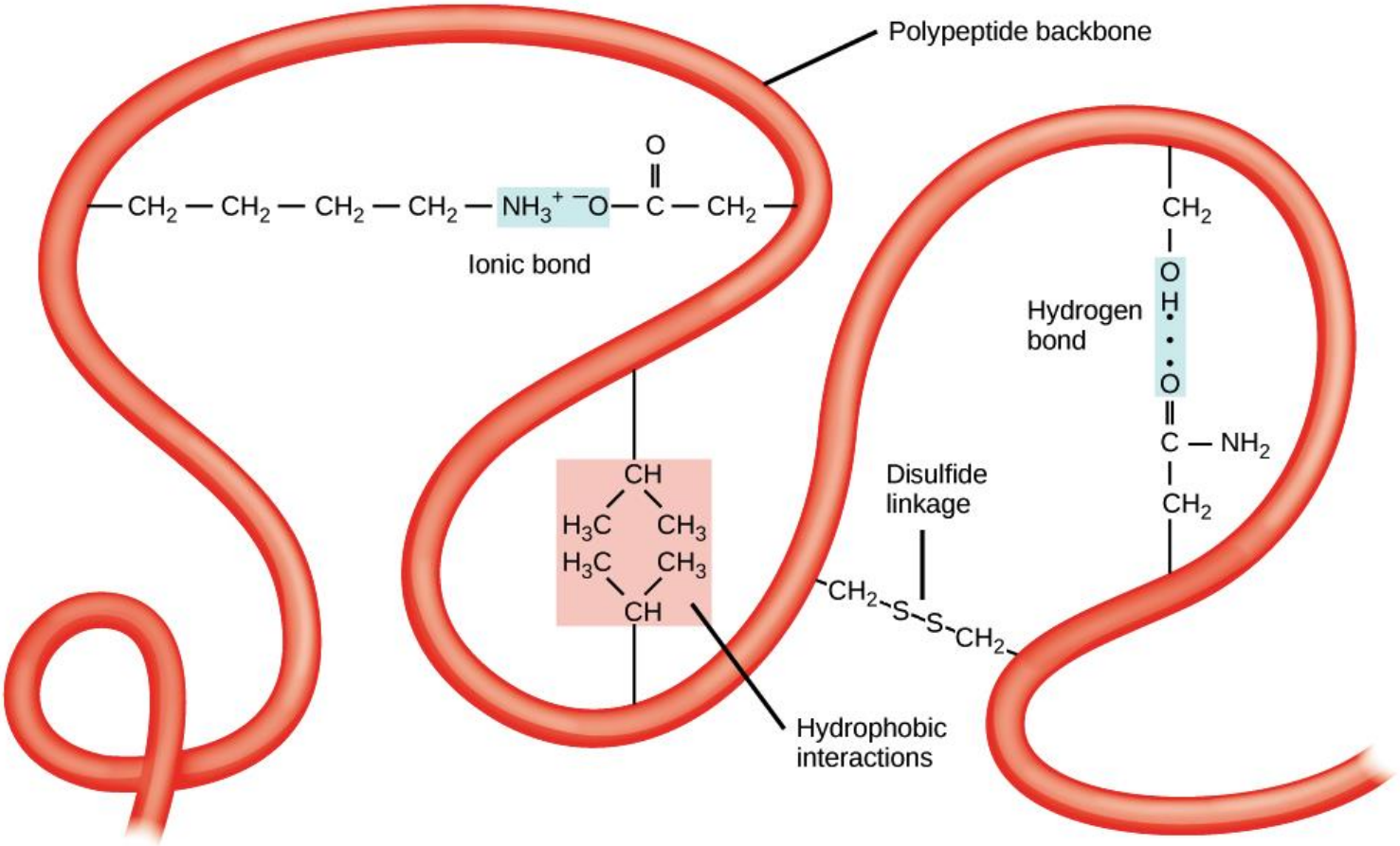


Image modified from OpenStax Biology.

Denaturation and Protein folding

When a protein loses its higher-order structure, but not its primary sequence, it is said to be **denatured**

Factors: temperature, pH….

Denatured proteins are usually non-functional



Amino acids are subunits of proteins

Responsible for thousands of distinct functions in cells

Regulation: control pr activity

Structure: tubulin – microtubules, histones- chromosomes

Signaling: monitoring of the environment and transmitting resultant information

Transport: flow of small molecules and ions across membrane

Catalysis: enzymes

Motor proteins: generation of force for movement

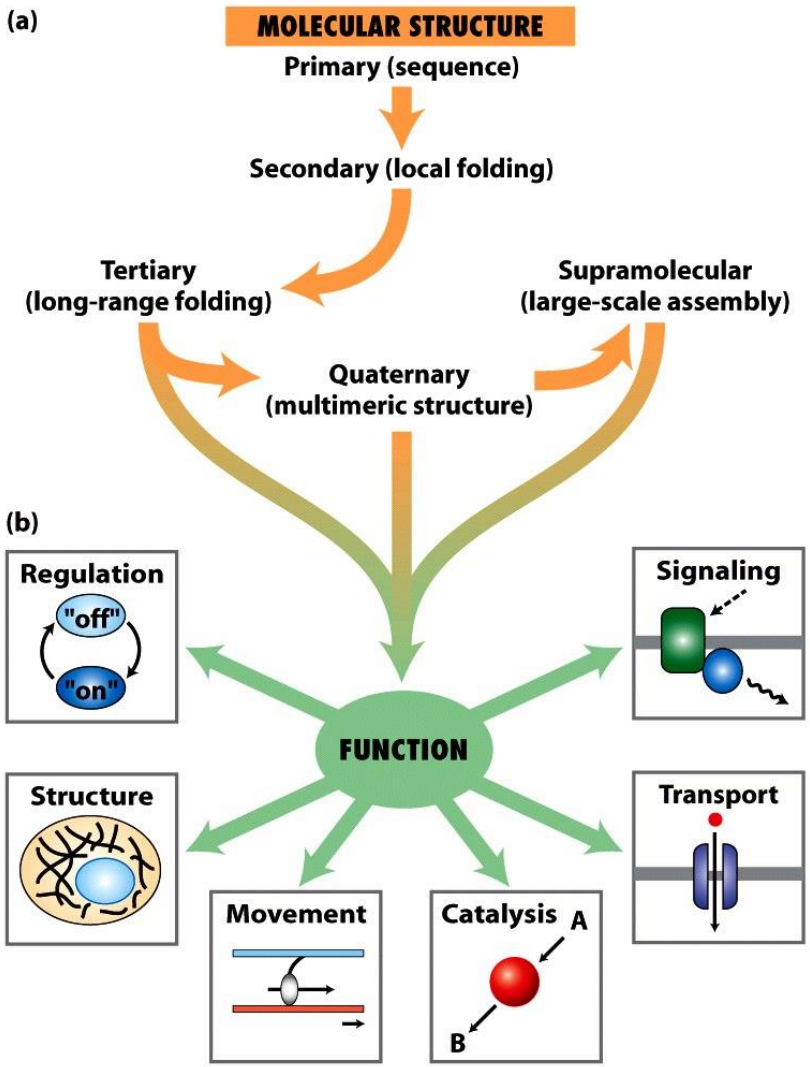


Figure 3-1
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Overview of protein structure and function

Enzyme

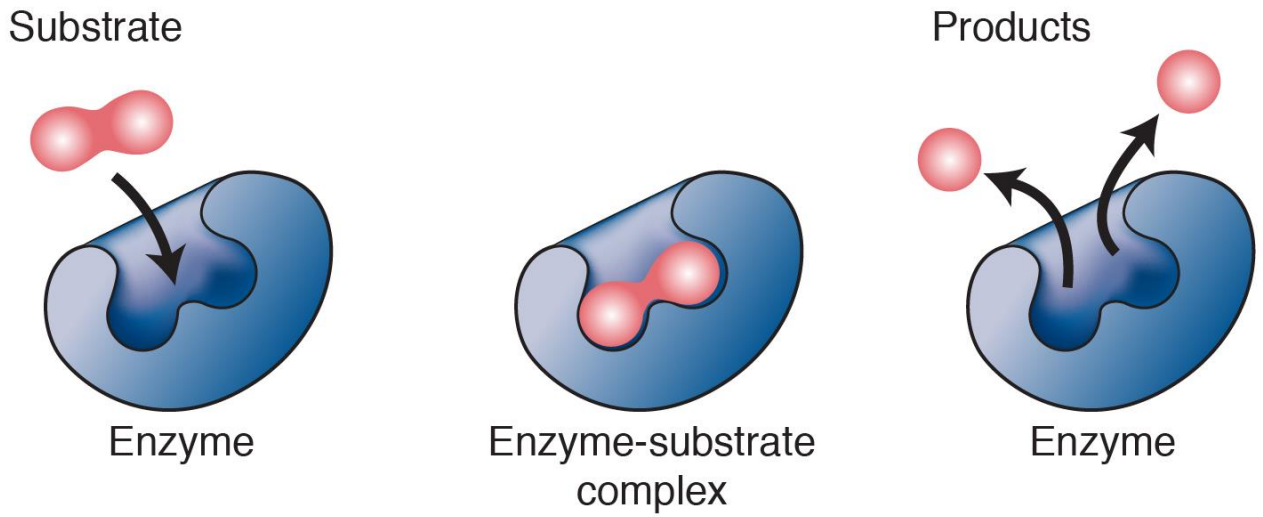
Help **speed up** metabolism, or the chemical reactions in our bodies and do **not chemically changed** at the end of the reaction

Name = substrate + ase

Characteristics:

- A small amount is needed to catalyze a lot of substrate.
- Specific – each class of enzymes will catalyse only one particular reaction.
- Enzymes catalyse reversible reactions
- Many enzymes are only able to work with in presence of a coenzymes (or cofactor).
- Effected by changes in temperature and pH

Mechanism of enzyme activity



NIH, 10/2022

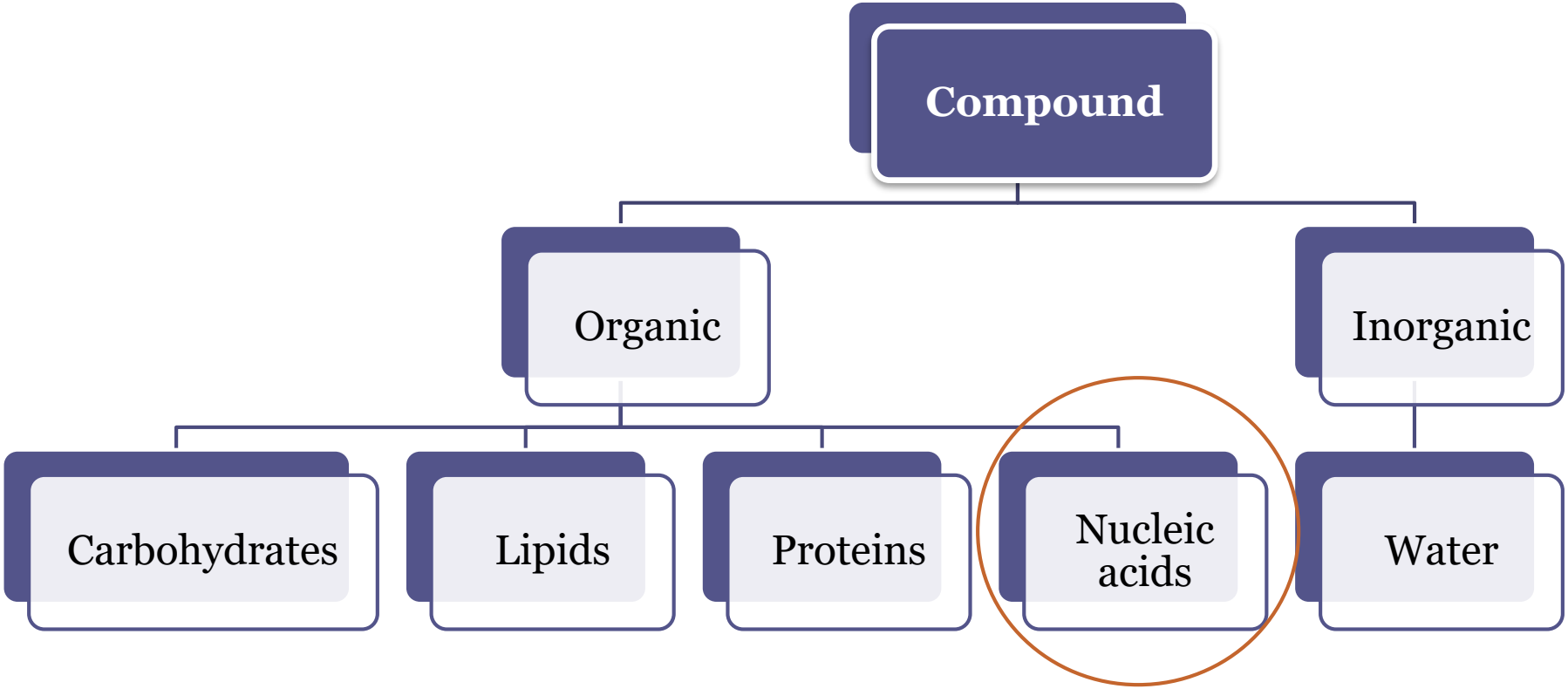
Factors affect the activity of enzymes:

1. pH

2. Temperature

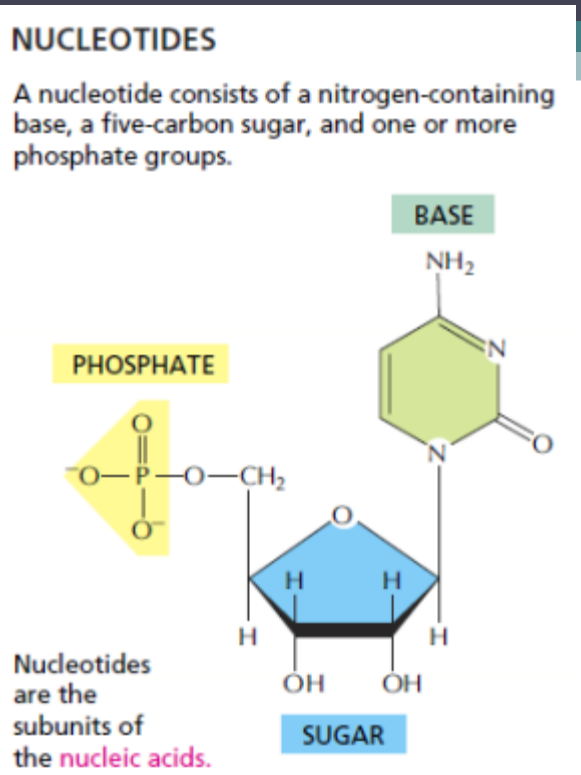
3. Concentration of enzyme

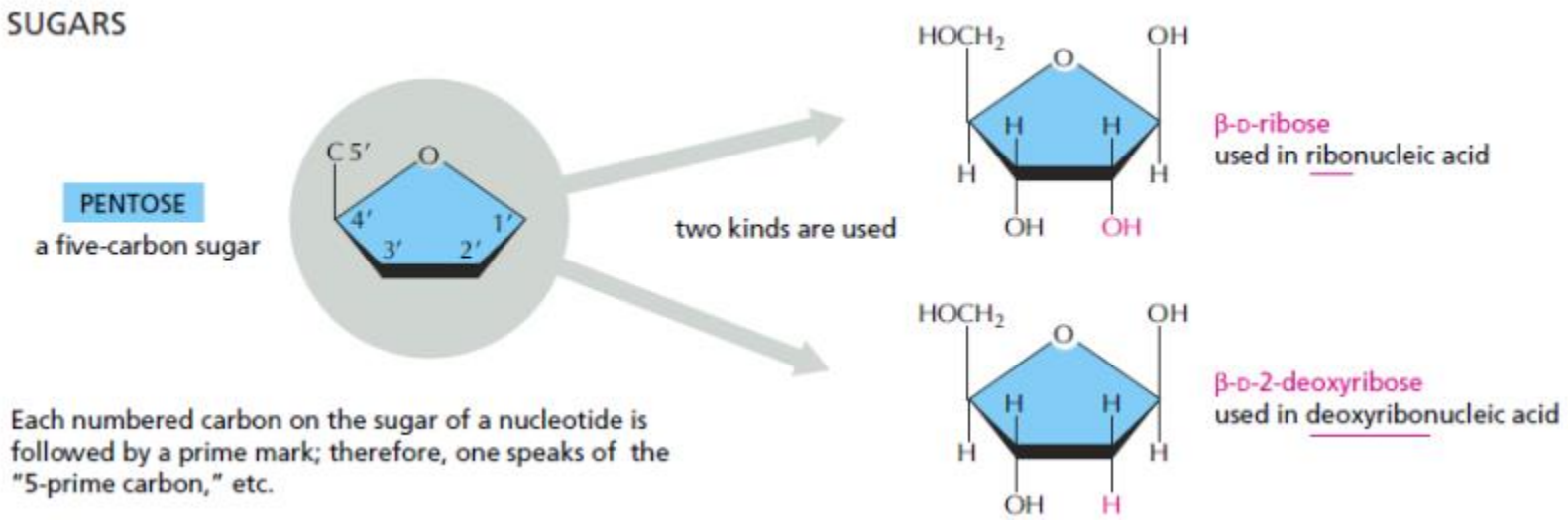
4. Concentration of substrate


Compound
Organic
Inorganic
Carbohydrates
Lipids
Proteins
Nucleic acids
Water

Nucleotides

- ❖ Pentose sugar
  - ribose → ribonucleotides
  - deoxyribose → deoxyribonucleotides
- ❖ One or more phosphate groups
- ❖ Bases





Panel 2-6 *Molecular Biology of the Cell* (© Garland Science 2008)

Bases

pyrimidines

–all derive from

six-membered pyrimidine ring

- cytosine (C), thymine (T) and uracil (U)

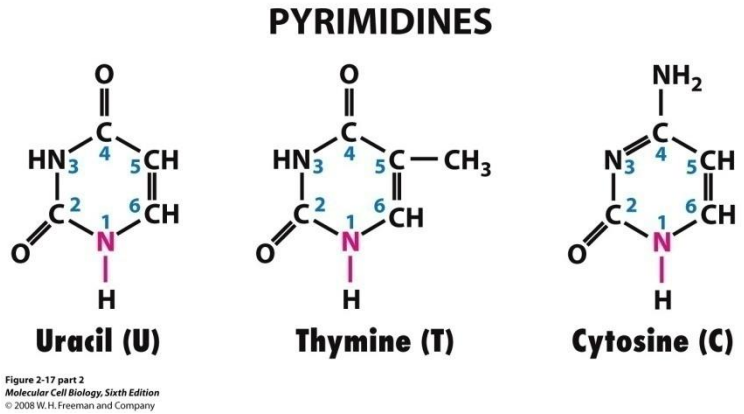


Figure 2-17 part 2
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purines

–have a second, five membered ring fused to the

six-membered ring

- adenine (A) and guanine (G)

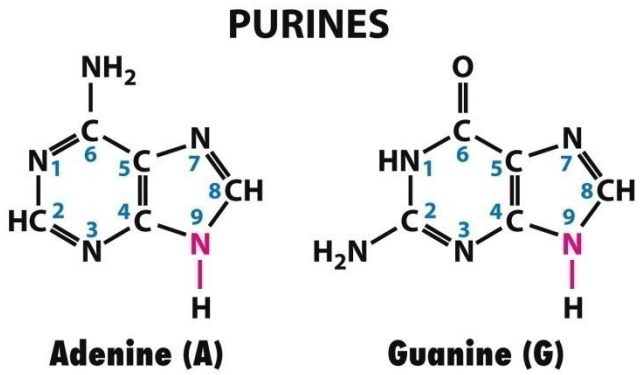


Figure 2-17 part 1
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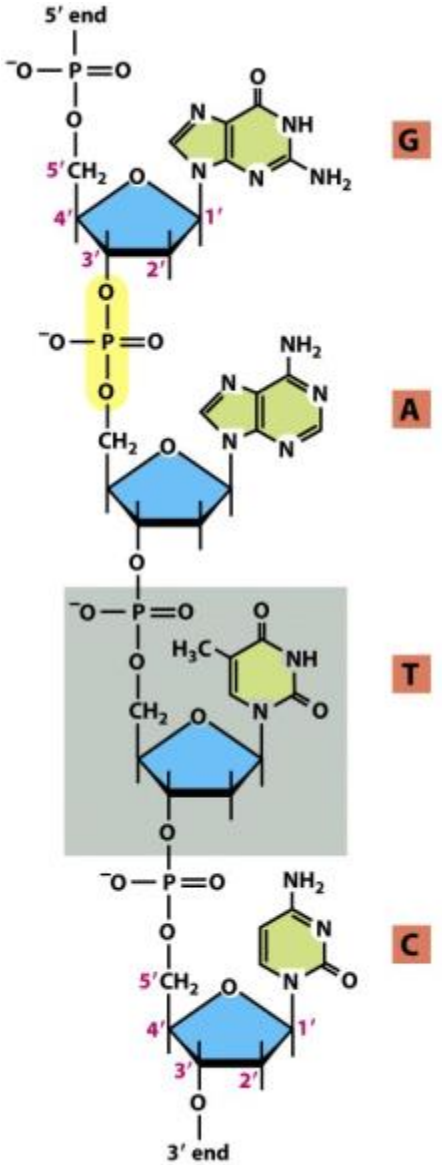


Figure 2-28 Molecular Biology of the Cell 5/e (© Garlan

❖The most important role → storage of biological information

❖ Two types of nucleic acids:

- **RNA** – ribose + A, G, C i U; mostly single-stranded
- **DNA** – deoxyribose + A, G, C i T; double stranded helix

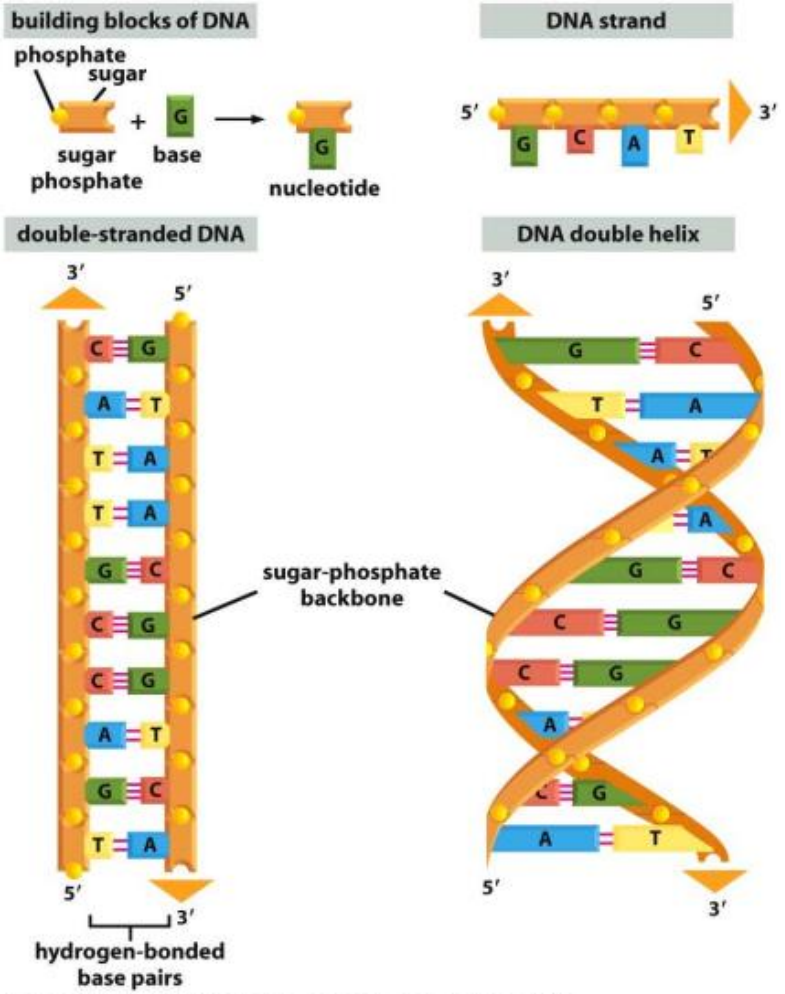


Figure 4-3 Molecular Biology of the Cell 5/e (© Garland Science 2008)

DNA and its building blocks

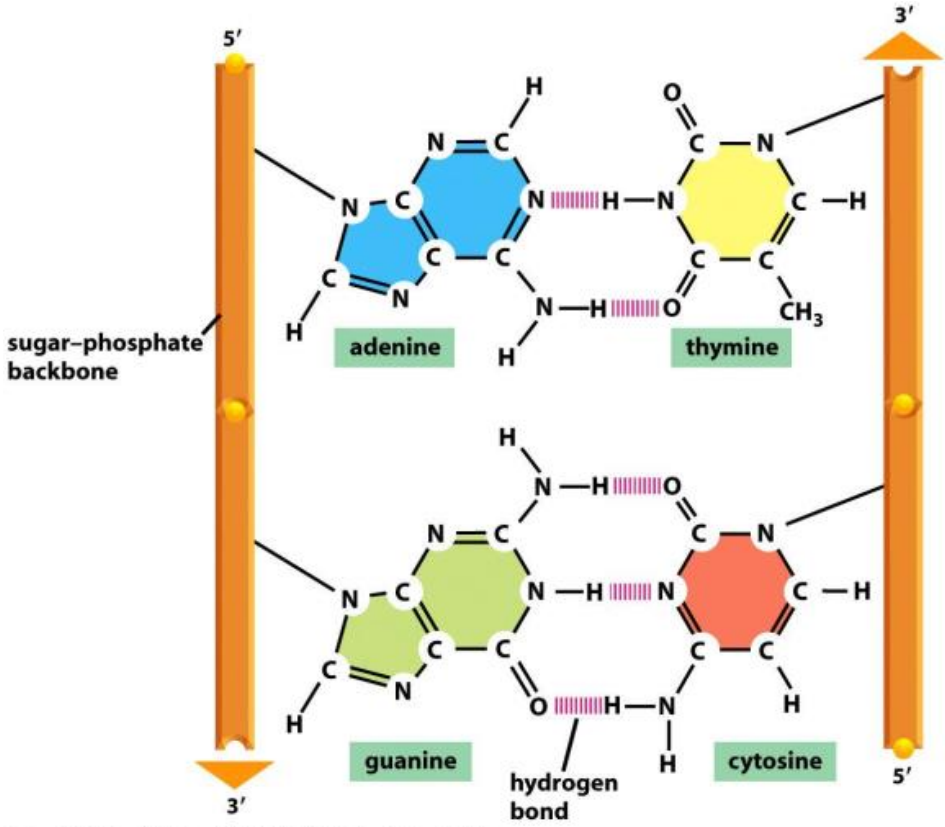


Figure 4-4 Molecular Biology of the Cell 5/e (© Garland Science 2008)

Complementary base pairs in the DNA double helix

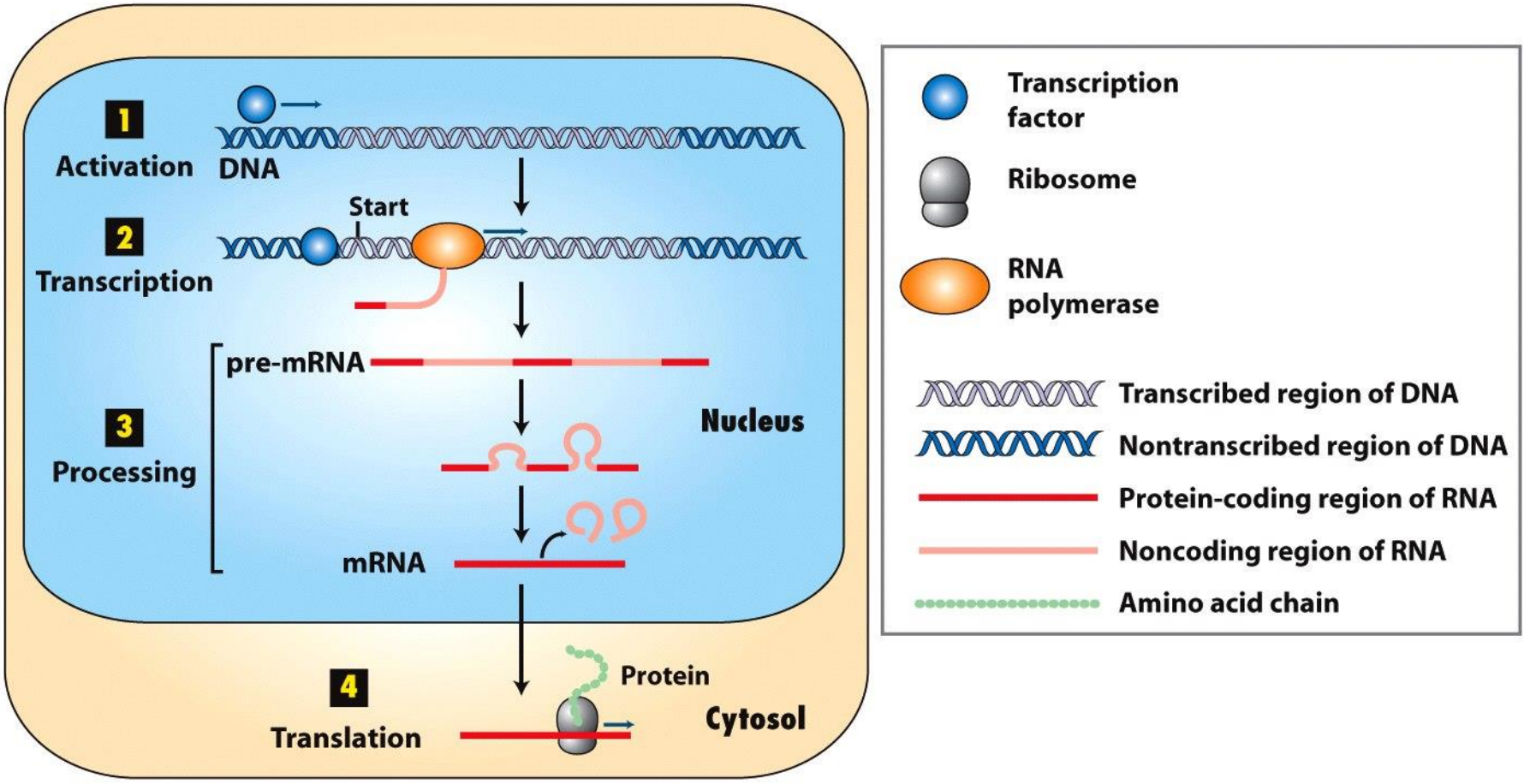
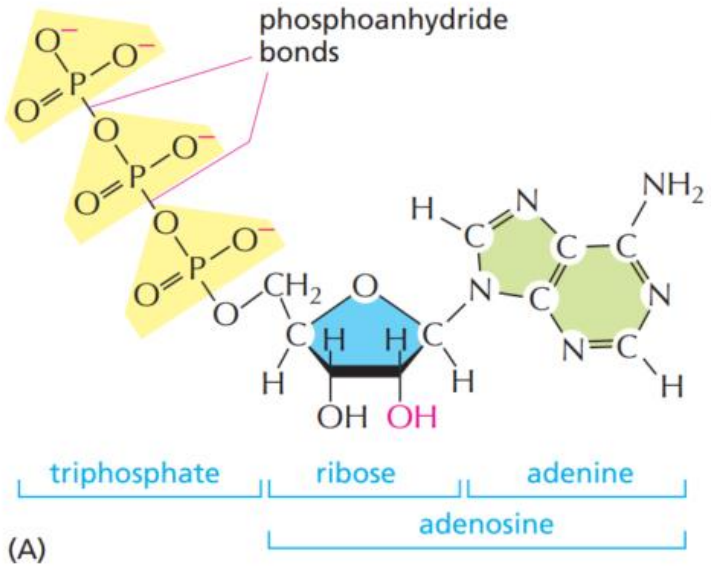


Figure 1-11
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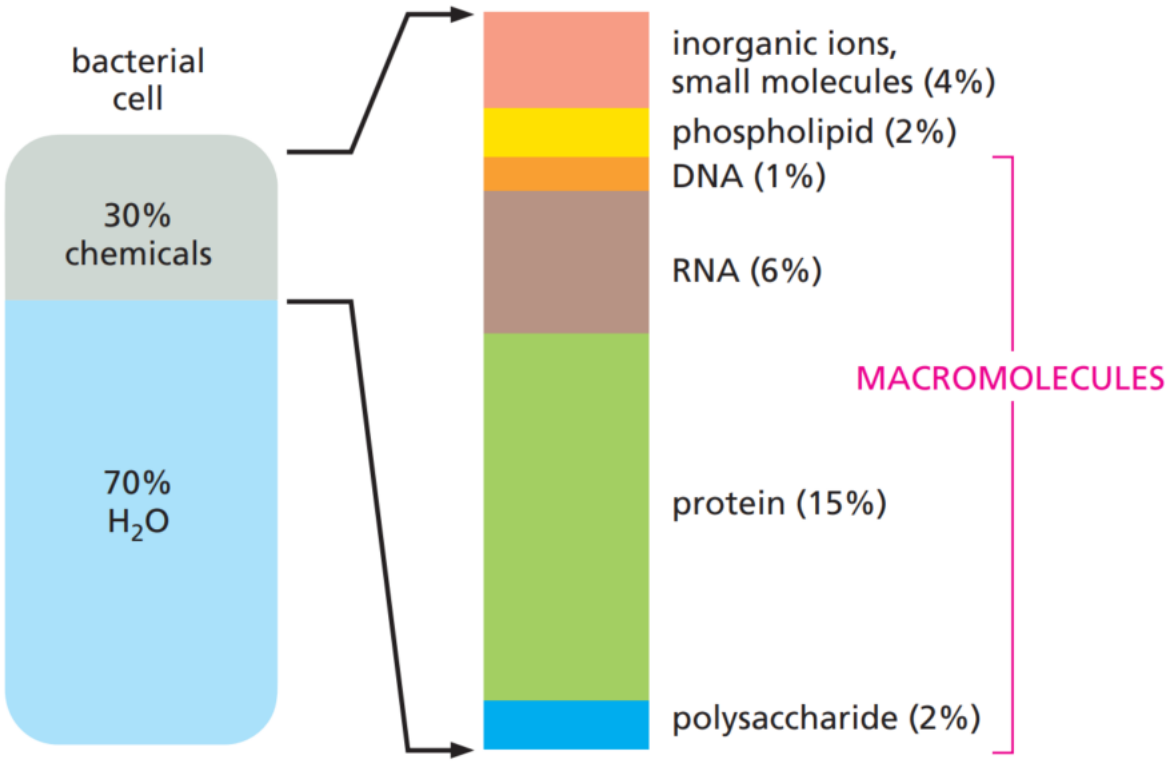
The coded information in DNA is converted into the amino acid sequences of protein by a multistep



Adenosine triphosphate (ATP) is a crucially important energy carrier in cells. (a) Structural formula, in which the three phosphate groups are shaded in *yellow*.

Biological molecules





Macromolecules are abundant in cells.
The approximate composition (by mass) of a bacterial cell is shown. The composition of an animal cell is similar.

Molecular complementarity

Molecular complementarity is the lock-and-key fit between molecules whose shapes, charges, and other physical properties are complementary.

Multiple noncovalent interactions can form between complementary molecules, causing them to **bind tightly**.

Intramolecular complementarity establishes the 3D shapes of proteins

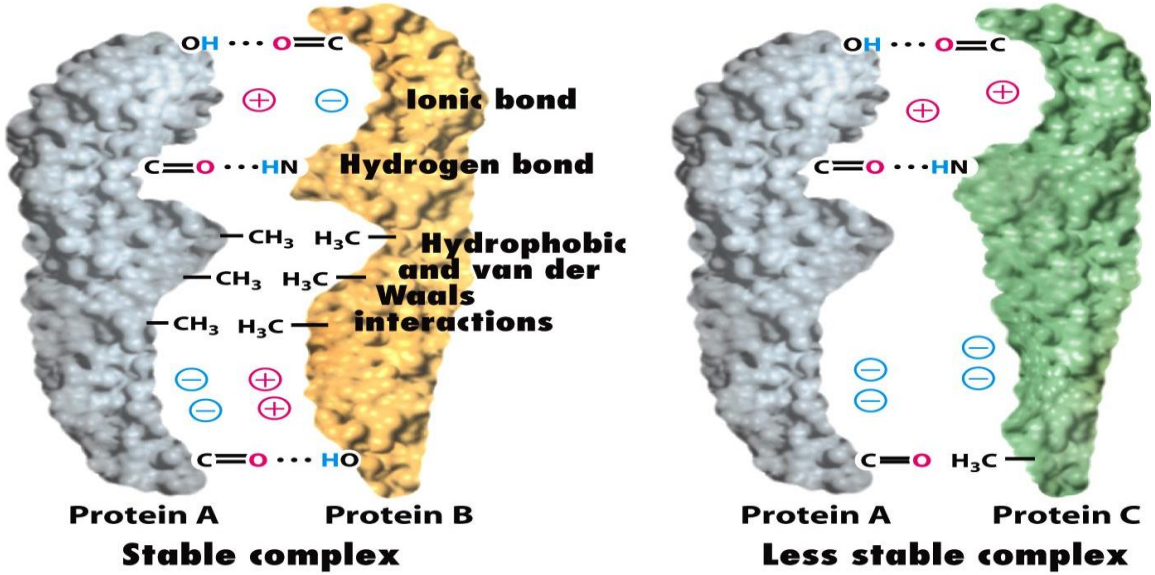


Figure 2-12
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